



Tick size, liquidity for small and large orders, and price informativeness: Evidence from the Tick Size Pilot Program[☆]

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ARTICLE INFO

Article history:

Received 10 October 2018

Revised 23 April 2019

Accepted 18 May 2019

Available online 26 November 2019

JEL classification:

G10

G14

G18

Keywords:

Liquidity

Tick size

Pilot program

Pricing efficiency

Liquidity spillover

ABSTRACT

Using limit order books across all US exchanges, we show that while liquidity for small orders (e.g., the quoted and effective spreads) decreases, liquidity for large orders (e.g., the cumulative depth and the price impact of multiple trades) improves after the implementation of the Tick Size Pilot Program. We find significant spillover effects on liquidity for small and large orders that extend beyond the top of the book. Finally, we show that the pilot program results in an improvement in pricing efficiency, an increase in trade size, and a decrease in the number of trades.

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1. Introduction

On October 3, 2016, the US Securities and Exchange Commission (SEC) launched the Tick Size Pilot Program to assess the impact of an increase in tick sizes on the market quality of small and illiquid stocks.¹ In this paper, we analyze the effect of this regulatory experimental shock on the

execution cost of small and large orders. Our main question is whether the larger tick size, which will most likely widen inside spreads and thus increase the cost of immediacy for smaller orders, increases the execution cost of large orders. We address this question by looking at consolidated liquidity at multiple price steps across all US exchanges relative to other studies that look at liquidity only

[☆] This work is supported in part by [National Science Foundation ACL-1541215](#). We thank OneMarket Data for the use of its OneTick software. We thank the editor (Bill Schwert) and an anonymous referee for many valuable comments and suggestions that greatly improved the paper.

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¹ In March 2011, the Treasury Department organized a conference to solicit ideas on how to close “The U.S. listing gap” (Doidge, Karolyi, and Stulz, 2017), a phenomenon that proportionally fewer US companies are listed on stock exchanges than companies in other countries. Conference participants, the IPO Task Force, concluded that this crisis is mainly the result of three factors: (1) costs for going public, (2) lack of information about emerging growth companies, and (3) a shift in trading in-

terest away from long-term investing in emerging growth companies toward high-frequency trading of large-cap stocks (IPO Task Force, 2011, pp. 1–2). In response, Congress passed the Jumpstart Our Business Startups Act (JOBS Act) in 2012. The JOBS Act defines emerging growth companies as companies that went public less than five years ago, have revenues less than \$1 billion in 2012 dollars, issued less than \$1 billion of non-convertible debt within a three-year period, and are not deemed to be a large accelerated filer, and establishes regulatory relief for these companies, thereby addressing the first concern of the IPO Task Force. The JOBS Act also addresses the second and third concerns by providing the option to amend Rule 612 of Regulation NMS, which, for stocks trading above \$1, set the minimum (quoted) price variation (the tick size) to one cent, in the hope that a larger tick size may improve trading and information environments for these companies.

at the top of the book or the depth of the book in one market. Equally important, we examine possible mechanisms and channels through which the pilot program affects liquidity by linking changes in liquidity for small and large orders to the three rules implemented in the pilot program. Finally, we investigate potential consequences of changes in the liquidity for small (retail, likely noise) and large (institutional, potentially informed) orders on information flow and pricing efficiency.

A change in tick sizes has a profound impact on the expected profits of liquidity providers. In a limit order market in which the tick size is zero, competition between liquidity providers leads to zero profit (Anshuman and Kalay, 1998). If the tick size is a binding constraint on spreads, liquidity providers cannot compete by offering better prices for small trades and, therefore, the binding tick size provides a wealth transfer from liquidity demanders to liquidity providers (Anshuman and Kalay, 1998; Goettler et al., 2005), from retail traders to institutions (Seppi, 1997), and from liquid to illiquid stocks (Werner et al., 2015). “Amending the tick size in a limit order market,” Goettler et al. (2005) conclude, “primarily perturbs the division of the spoils.” That the tick size merely allocates trading gains is largely a result of traders endogenously choosing between providing and demanding liquidity. Because a market can function only with traders who demand liquidity and others who provide it, a tick size that is too small or too large can lead to market failures (Seppi, 1997; Portniaguina et al., 2006). Therefore, tick sizes are important not only for allocating trading gains but also for welfare.

The pilot program stocks are divided into three test groups and one control group. The pilot stocks assigned to Test Group One (TG1) are quoted in \$0.05 increments but traded in \$0.01 increments. The pilot stocks assigned to Test Group Two (TG2) are quoted and traded in \$0.05 increments. The pilot stocks assigned to Test Group Three (TG3) are quoted and traded in \$0.05 increments and are also subject to the trade-at rule. The trade-at rule, in general, prevents exchanges from executing a trade at the national best bid and offer (NBBO) without also displaying the NBBO, giving liquidity providers an incentive to display instead of hide their trading interests. The pilot stocks in the control group are quoted and traded in \$0.01 increments or smaller. The quote (Q) rule applies to TG1, TG2, and TG3, the trade (T) rule applies to TG2 and TG3, and the trade-at (TA) rule applies to only TG3.

We collect data for all stocks in the pilot program from 2015 to 2017. We obtain trade and quote data from the Trade and Quote (TAQ) database and the ten best bid and ask prices and sizes in the consolidated limit order book constructed from the Thomson Reuters Tick History (TRTH) database. Most of our tests are based on a difference-in-differences method, in which we compare the effect on the pilot stocks with the effect on the control stocks before and after the event month (October 2016), the month in which the tick size increased for the pilot stocks.²

To assess the effect of the pilot program on the redistribution of trading costs, we examine whether the program increases the execution cost of small orders and decreases the execution cost of large orders. We measure the execution cost of small orders by the quoted and effective spreads and the price impact of a trade. To assess the execution cost of large orders, we consider both the cumulative depth and the percentage of time during which at least five thousand shares are available within 20 cents or 5% of the NBBO in the limit order book.³ We also calculate the cost of a round-trip trade of five hundred and five thousand shares using the ten national best bid and ask prices and sizes and the cumulative price impact of multiple trades.

We find that the pilot program increases the execution cost of small orders that demand immediacy. For the pilot stocks, both the quoted and effective spreads increase, on average, by 15 basis points, which is statistically and economically significant. Although the price impact of a trade increases, the realized spread also increases for the pilot stocks, indicating an increase in liquidity providers' revenues. As to the execution cost of large orders, the cumulative depth doubles, the availability of trading five thousand shares triples, the cost of a round-trip trade of five hundred and five thousand shares decreases, and the cumulative price impact of multiple trades decreases. Using Rule 605 data, we further show that the cost of immediacy decreases for large orders (above two thousand shares). Consistent with the decrease in liquidity for small orders and the increase in liquidity for large orders, the number of trades decreases and the average trade size increases.

To shed light on underlying mechanisms and channels through which the pilot program affects liquidity, we develop our hypotheses regarding whether and how the quote (Q), trade (T), and trade-at (TA) rules could affect liquidity and the cost of trading by exploring the implications of these rules. Empirical results are generally consistent with these implications. For example, the Q rule results in a greater increase in quoted, effective, and realized spreads when the larger quote tick size is more likely a binding constraint (TBC) on spreads. The T rule increases effective spreads because it decreases the probability of price improvements. The TA rule reduces effective spreads because it increases visible liquidity. In a similar vein, we show that the Q rule increases the depth and decreases the execution cost of large orders and that the increase and decrease are greater for stocks with larger TBC values.

Rindi and Werner (2017) show that the pilot program has a spillover effect; i.e., it also affects the liquidity of stocks in the control group. Rindi and Werner (2017) analyze the spillover effect using the NBBO quotes, which capture the spillover effect for small orders. We extend their study by finding evidence of liquidity migration from the control stocks to the pilot stocks for large orders. We also show that the spillover effect of the pilot program extends beyond the top of the book liquidity.

² Boehmer, Jones, and Zhang (2020) provide evidence that the difference-in-differences approach can underestimate or overes-

timate the effect of a regulatory change, depending on whether direct and indirect (i.e., spillover) effects are in the same or opposite directions.

³ We find similar results when we use the cumulative depth available at the ten best bid and ask prices provided in the TRTH database.

Although liquidity is in general desirable and often linked to more efficient prices (Chordia et al., 2008), the very nature of impounding information into prices is also one of the main reasons that assets are illiquid. That is, liquidity providers face adverse selection risk, the risk of trading with informed investors (Glosten and Milgrom, 1985). How pricing efficiency affects optimal tick sizes is modeled by, for example, Anshuman and Kalay (1998), who conclude that the optimal tick size is decreasing in the amount of private information (see their Proposition 3). Considering that at least some market participants are concerned about the lack of information in small stocks, we investigate whether an increase in the tick size improves “the availability and flow of information for investors” (IPO Task Force, 2011, p. 2).

We use fundamental volatility, which is the difference between the volatility of quote midpoint returns and the volatility of pricing error (Hasbrouck, 1993; Boehmer and Kelley, 2009), as a proxy for how much information is incorporated into prices. We employ a measure of firm-specific information based on price non-synchronicity first proposed by Roll (1988) and used in Chen et al. (2007) and Morck et al. (2013). We also test whether the larger tick size affects market efficiency by analyzing its effect on the return predictability of past order flow changes (Chordia et al., 2008). We find that, for the pilot stocks, fundamental volatility increases and prices reflect more stock-specific information. We also find that return predictability decreases (i.e., pricing efficiency increases), confirming the conjecture that informed traders can profit more from their private information. Combined with the lower expected trading cost of large orders, these results underscore the importance of the liquidity for large trades for information production. In addition, the increase in the execution cost of small orders could deter noise trading (i.e., retail traders), thereby also improving pricing efficiency.

A rich empirical literature exists on how tick sizes affect market quality, which is summarized in an excellent survey article by the SEC (2012). We are aware of nine contemporaneous papers investigating the Tick Size Pilot Program. Of these papers, three are only indirectly associated with our study as they focus on how tick sizes, exchange fees, and trading volume are related (Cox et al., 2017; Lin et al., 2017; Comerton-Forde et al., 2019). Comerton-Forde et al. (2019) argue that by offering traders sub-tick price improvements, inverted fee venues enhance competition for liquidity provision and increase information impounded into prices through limit orders. The closest papers to ours are Rindi and Werner (2017), Penalva and Tapia (2017), and Griffith and Roseman (2019), who study the impact of the tick size on market liquidity, Whitby and Blau (2018), who analyze the impact on trading activity and return volatility, Albuquerque et al. (2017), who consider pricing efficiency, and Hu et al. (2018), who analyze the impact on market quality as measured by spreads, depths, trading volume, volatility, and price efficiency.

Most research points out negative consequences of the larger tick size. For example, Griffith and Roseman (2019, p. 104) find that “the tick size increase is binding and leads to a significant decrease in liquidity in the limit order book.” Hu et al. (2018, Abstract) summarize their finding as “Over-

all, we find that on average, relative to stocks in the Control Group, market quality deteriorates for stocks in the Test Groups.” Rindi and Werner (2017, Abstract) show that the larger tick size increases quoted and effective spreads, raising “the cost for retail-sized liquidity demanding orders by almost fifty percent.” Rindi and Werner (2017, Abstract) find that the larger tick size results in an increase in trade sizes and conjecture that “institutions may benefit from the deeper quotes.” Rindi and Werner (2017) do not provide evidence on the conjecture because their study focuses on NBBO quotes (spreads and depths).

We contribute to the literature in several ways. First, to the best of our knowledge, we break ground in analyzing the effects of the Q , T , and TA rules on liquidity using cumulative depths across all US exchanges and providing evidence that the effect of the pilot program extends beyond the top of the book. Both Penalva and Tapia (2017) and Griffith and Roseman (2019) have access to cumulative depth on Nasdaq only (from Total-View-ITCH data). Our result that liquidity increases even for a modest trade size of five hundred shares is in contrast to Griffith and Roseman (2019), underscoring the importance of having access across all exchanges.⁴ Second, our study provides in-depth analyses of the effects of the quote, trade, and trade-at rules on liquidity for small and large orders by exploring five distinct implications of these rules. As a result, our study not only provides evidence on the effect of the larger tick size on liquidity but also sheds some light on potential mechanisms and channels through which the pilot program affects liquidity. Third, we provide evidence of liquidity spillovers between the pilot and control stocks for both small and large orders. Fourth, we offer evidence on how the tick size affects information flow and pricing efficiency. Although Albuquerque et al. (2017) analyze the effect of the pilot program on pricing efficiency, they do not explore the pricing effect of the T , Q , and TA rules. Whitby and Blau (2018) analyze the impact of the pilot program on trading activity and return volatility using daily data, providing only limited insights on the impact.

The rest of the paper proceeds as follows. In Section 2, we develop our hypotheses regarding the effects of the quote, trade, and trade-at rules on various liquidity measures. Section 3 explains data sources, sample construction, and variable measurement methods. Section 4 provides evidence regarding the aggregate effect of the tick pilot program on our liquidity measures. Sections 5 presents empirical findings on our hypotheses developed in Section 2. Section 6 explores the effect of the pilot program on the control stocks, whether the effect of the pilot program extends beyond the top of the book, and whether our results are driven by phantom liquidity. Section 7 analyzes the effects of the pilot program on information flow, price informativeness, and trading activity. Section 8 ends the paper with a brief summary and concluding remarks.

⁴ The importance of observing depth across all exchanges could be explained by the fact that substantial trading diverted to inverted exchanges (see Cox, Van Ness, and Van Ness, 2017; Lin, Swan, and Mollica, 2017; Comerton-Forde, Gregoire, and Zhong, 2019).

2. Hypotheses development

To help develop hypotheses on how *Q*, *T*, and *TA* rules can affect different dimensions of liquidity, we first postulate the following five implications of the rules.

- (11) The *Q* rule increases the probability that the quote tick size is a binding constraint (TBC) on the inside quoted spread.
- (12) The *Q* rule results in coarser quotable prices.
- (13) The *T* rule decreases front running because a larger trade tick size imposes a larger cost on front runners (with a larger trade tick size, front runners need to improve the price by a larger magnitude to trade ahead of other suppliers of liquidity based on price precedence).
- (14) The *T* rule decreases the probability of price improvements because liquidity providers are less likely to offer price improvements when the size of the minimum possible price improvement is larger. For instance, liquidity providers who are willing to provide 2-cent price improvements are unlikely to provide a price improvement at all if the minimum possible price improvement they could offer is 5 cents due to the *T* rule. Implication 14 could also follow from a decrease in hidden liquidity that results from the *T* rule. The *T* rule can lower investors' desire to hide trading interests (e.g., using hidden limit orders) to the extent that it reduces their concerns about front running (i.e., implication 13). All things being equal, a reduction in hidden liquidity (or an increase in visible liquidity) implies smaller room for price improvements because it implies a reduction in quoted spreads.
- (15) The *TA* rule increases visible liquidity by moving hidden liquidity (e.g., from hidden orders or dark trading venues) to visible liquidity because, under the *TA* rule, a venue can execute incoming market orders at the NBBO only if the best bid or ask was also displayed at the venue. Comerton-Forde et al. (2019) find a decrease in dark trading by over 5% for stocks in Test Group Three (TG3) after the imposition of the *TA* rule.

Based on these five implications, we now state our hypotheses on the effects of the *Q*, *T*, and *TA* rules on liquidity measures.

Hypothesis 1 (the cost of immediacy: the quoted and effective spreads). *Implication 11 predicts that the Q rule increases the quoted and effective spreads for those stocks for which the new tick size (\$0.05) is a binding constraint on the quoted spread and that this increase is greater for stocks with larger TBC. From implication 14 and the definition of the effective spread (i.e., the quoted spread – price improvement), the T rule should increase the effective spread. Implication 15 predicts that the TA rule decreases the quoted and effective spreads.*⁵

⁵ From implication 15, we expect the *TA* rule to decrease the quoted spread by increasing displayed liquidity. An increase in displayed liquidity implies smaller room for price improvements because it implies a

Cordella and Foucault (1999) suggest that a firm's share price moves toward its intrinsic value more quickly when the quote tick size is larger to the extent that it provides liquidity suppliers with economic rents (i.e., actual spreads are larger than the equilibrium spreads that would prevail if the tick size were not a binding constraint on spreads). Hence, the price impact of a trade increases with the quote tick size, especially when the quote tick size is a binding constraint on spreads (implication 11). Front running reduces the incentive to collect and analyze information and to use it in trading. To the extent that the larger trade tick size reduces front running (implication 13), traders have greater incentives to gather and trade on private information when the trade tick size is larger, increasing the (permanent) price impact of a trade. The realized spread, which measures the market-making revenue, is the difference between the effective spread and the price impact of a trade. Combined with Hypothesis 1, which applies to the effective spread, these considerations lead to Hypothesis 2.

Hypothesis 2 (the price impact of a trade and the realized spread). *The Q and T rules increase the price impact of a trade, and the effect of the Q rule is greater for stocks with larger TBC. Effects of the Q and T rules on the realized spread depend on the relative magnitudes of these effects on the effective spread and the price impact of a trade. The TA rule decreases the realized spread.*

The *Q*, *T*, and *TA* rules can increase depths at the NBBO through different channels. The larger quote tick size could increase depths at the NBBO simply because it implies coarser quotable prices (i.e., implication 12). The larger quote tick size also could increase depths at the NBBO to the extent it increases the likelihood that the tick size is a binding constraint on spreads in the pilot period (i.e., implication 11). Liquidity providers would increase depths when they are forced to quote wider spreads than equilibrium spreads due to the larger tick size if they are to provide the same level of liquidity. Liquidity providers can increase depths even more to the extent that larger spreads imply more attractive prices (lower bid and higher ask prices). These increases would be positively related to *TBC*. Traders are less likely to offer liquidity when confronted with the prospect of front running and the higher transaction costs it imposes on them (Harris, 2003). Hence, the *T* rule is likely to increase depths to the extent that the larger trade tick size decreases front running by increasing the cost of front running (implication 13). From implication 15, we expect that the *TA* rule increases depths at the NBBO.

Goldstein and Kavajecz (2000) show that a decrease in tick sizes results in a reduction in depths throughout the limit order book, suggesting that the *Q* rule could increase both the cumulative depth and the percentage of time during which a large number of shares are available at the NBBO and nearby quotes. A possible link between the *Q* rule and these liquidity measures for large orders is that a

decrease in the quoted spread. The effect of the *TA* rule on the quoted spread is likely to be greater than the effect of the *TA* rule on price improvement because the latter effect is conditional on the former effect. Hence, we expect the *TA* rule to decrease the effective spread.

larger quote tick size could increase order sizes, which, in turn, gives liquidity providers an incentive to quote larger sizes, increasing both liquidity measures for large orders. Because the NBBO depth is a part of the cumulative depth, we expect that the positive effect of the *Q* rule on the cumulative depth and the percentage of time with large depths is greater for stocks with larger *TBC* values. The *T* rule is likely to increase liquidity measures for large orders to the extent that it decreases front running (implication I3). The *TA* rule is likely to increase them to the extent that it increases visible liquidity (implication I5). These considerations lead to Hypothesis 3.

Hypothesis 3 (the NBBO depth, the cumulative depth, and the percentage of time with large depths). *The Q rule increases the NBBO depth, the cumulative depth, and the percentage of time during which a large number of shares are available at the NBBO and nearby quotes, and these increases are greater for stocks with larger TBC values. The T and TA rules also increase these liquidity variables.*

All things being equal, the increase in the cumulative depth and the percentage of time with large depths in Hypothesis 3 implies a decrease in the execution cost of large orders, and this decrease is greater for stocks with larger *TBC*. A large trader, who intends to buy or sell a large number of shares, can submit a large order or a number of small child orders over time. Regardless of whether the large trader submits a large order or multiple child orders, the increase in the cumulative depth and the percentage of time with large depths in Hypothesis 3 implies a decrease in the cumulative price impact of trades during a given time period, and this decrease is likely greater for stocks with larger *TBC*. The effects of the *T* and *TA* rules on these variables are likely similar to those on the cumulative depth. These considerations lead to Hypothesis 4.

Hypothesis 4 (the execution cost of larger orders and the cumulative price impact of multiple trades). *The Q rule decreases the cost of a round-trip trade for a given size and the cumulative price impact of trades during a given time period, and these decreases are greater for stocks with larger TBC values. The T and TA rules decrease both liquidity measures.*

Table OA1 in the Online Appendix provides a summary of these hypotheses, together with how they are related to the *Q*, *T*, and *TA* rules through their key implications. The Online Appendix is provided in the Social Science Research Network version of this paper.

3. Data sources, sample construction, and variable measurement

This section explains our data sources, sample selection procedures, and variable measurement methods.

3.1. Data sources and sample construction

On October 3, 2016, the SEC launched the Tick Size Pilot Program to study the impact of an increase in tick sizes on market quality. The SEC identified eligible stocks that trade

above \$2 and have a market capitalization of below \$3 billion. Following a stratified random sampling process, eligible stocks are separated into three pilot test groups (TG1, TG2, and TG3), each consisting of four hundred stocks, and a control group of approximately 1200 stocks. We obtain the list of 2395 stocks in the pilot program from the Financial Industry Regulatory Authority website.⁶ Following Rindi and Werner (2017), we drop test stocks (e.g., stocks with a name that looks something like “NYSE Tick Pilot Test Sym-Control”) (12 stocks), stocks that changed their group (e.g., from TG1 to the control group) or primary listing during the program (63 stocks), stocks added after the program initialization or deleted before the end of sample period (159 stocks), and stock-days with fewer than 20 trades. If more than half of a stock’s daily observations have fewer than 20 trades, all of its observations are dropped (167 additional stocks dropped). In addition, we keep only stocks with a share code of 10 or 11 in the Center for Research in Security Prices (CRSP) database (291 additional stocks dropped). After the screening processes, we have 1703 stocks left in our sample. Following Rindi and Werner (2017), we also drop the following stock-days: stock symbols NRCIA and PBIP on November 28, 2016; HUBS on August 4, 2016; VCRA on October 28, 2016; and DTLK on December 19–21, 2016. Unless otherwise noted, we define October 1, 2015 to September 23, 2016 as the pre-pilot period and October 31, 2016 to October 31, 2017 as the pilot period.

The Thomson Reuters Tick History (TRTH) database enables us to construct snapshots of limit order books from all US stock exchanges, which is not possible through the TAQ or Nasdaq TotalView-ITCH databases.⁷ As one of our goals in this paper is to examine the effects of the Tick Size Pilot Program on market quality beyond the NBBO, we use the TRTH database to estimate liquidity measures for large orders that require price and size (depth) information above the lowest ask price and below the highest bid price. Following Rindi and Werner (2017), we drop observations with a zero or negative ask or bid size and observations with a zero bid or ask price. We also delete stub quotes, which we define as quotes with a bid (ask) price more than 31.5% away from the national best bid (ask).⁸ The TRTH data set we use contains NYSE and Nasdaq stocks only.

We use the Daily TAQ database to compute the quoted spread, effective spread, realized spread, price impact, number of trades, trade volume, average trade size, return autocorrelation, return volatility, Hasbrouck measure, and return predictability of order imbalance. We use the Holden and Jacobsen (2014) filter when we construct the NBBO. Because the Daily TAQ data are time-stamped to the millisecond, we do not apply the interpolation algo-

⁶ See <http://www.finra.org/industry/tick-size-pilot-program>.

⁷ Several recent studies also use the TRTH database, including Lai, Ng, and Zhang (2014), Lau, Ng, and Zhang (2012), Marshall, Nguyen, and Visaltanachoti (2011), Kahraman and Tookes (2017), Gao, Lu, and Song (2019), and Jørgensen, Skjeltorp, and Ødegaard (2018).

⁸ According to an SEC regulation, a quote can be entered as far as 30% away from the NBBO, and an additional drift of up to 1.5% is allowed. See www.sec.gov/news/press/2010/2010-216.htm.

rithm suggested by [Holden and Jacobsen \(2014\)](#), which is designed to solve issues arising from multiple quotes or trades occurring in the same one-second interval. We classify each trade into a buyer- or seller-initiated trade using the method in [Ellis et al. \(2000\)](#). For both the TRTH and TAQ data, we use observations from 9:30 a.m. to 4:00 p.m.

3.2. Variable measurement

We measure the extent to which the new tick size is a binding constraint on quoted spreads (*TBC*) by the percentage of quoted spreads during the pre-pilot period that are equal to or smaller than the new quote tick size of 5 cents. To construct this measure, we calculate the quoted spread at each second during the trading day. Then, we compute the percentage of one second intervals during the day with quoted spreads less than or equal to 5 cents. (We replicate our empirical analyses using an alternative measure of *TBC*: the percentage of NBBO updates with quoted spread less than or equal to 5 cents. The results are similar.) We use the mean value of the daily *TBC* values for each stock during the pre-pilot period as our proxy for *TBC*.

The quoted spread at each NBBO update time t is computed as

$$\text{QuotedSpread}_t = \frac{(A_t - B_t)}{\text{Midquote}_t}, \quad (1)$$

where A_t is the national best ask price at time t , B_t is the national best bid price at time t , and Midquote_t is the quote midpoint at time t (i.e., the mean of A_t and B_t). We then obtain the daily quoted spread (QuotedSpread_d), which is the time-weighted mean value of QuotedSpread_t for each stock on each day. We measure the daily quoted depth in thousand's of US dollars (Depth_d) by the time-weighted mean value of the quoted depth at each NBBO update time for each stock on each day.

The effective spread for trade m on day d is measured by

$$\text{EffectiveSpread}_{m,d} = \frac{2 \times \text{BuySell}_m \times (\text{Price}_m - \text{Midquote}_t)}{\text{Midquote}_t}, \quad (2)$$

where Price_m is the transaction price, BuySell_m equals to 1 (−1) for buyer-initiated (seller-initiated) trades, and Midquote_t is the quote midpoint at the time of trade m . Each trade m 's $\text{EffectiveSpread}_{m,d}$ is multiplied by the number of shares executed for that particular trade and then divided by the total number of shares executed for day d . We sum all of the weighted $\text{EffectiveSpread}_{m,d}$ in day d to obtain the share-weighted EffectiveSpread_d . The effective spread captures the execution cost of a large order even if exchanges report trades and their sizes based on matched limit orders. For instance, if a market order of four thousand shares is matched with four resting limit orders (one thousand shares each) at different prices and reported as such, the difference between the mean value of these four prices and the mean quote midpoint correctly measures the execution cost of the order. We cannot infer the execution cost of large orders using the mean effective spread when there are both small and large orders unless we could isolate trades that are associated with large

orders. If the proportion of small orders is much higher than that of large orders in a given stock, the mean effective spread of the stock largely captures the trading cost of small orders.

We measure the price impact of trade m on day d by

$$\text{PriceImpact}_{m,d} = \frac{2 \times \text{BuySell}_m \times (\text{Midquote}_{t+\tau} - \text{Midquote}_t)}{\text{Midquote}_t}, \quad (3)$$

where τ is either one or five minutes. We then calculate the share-weighted PriceImpact_d for each day.

The realized spread for trade m on day d is computed as

$$\text{RealizedSpread}_{m,d} = \frac{2 \times \text{BuySell}_m \times (\text{Price}_m - \text{Midquote}_{t+\tau})}{\text{Midquote}_t}, \quad (4)$$

where τ is either one or five minutes. We find similar results when we use other time horizons (15, 30, 90, 120, 180, and 240 s). Again, we calculate the share-weighted RealizedSpread_d .

To help assess the trading cost of large orders, we estimate the time-weighted cumulative share depth available on both sides of the limit order book within 20 cents or 5% of the NBBO (*CumuDepth*). We also estimate the percentages of one-minute intervals during the day when five thousand shares are available on both sides of the book (*PAVAIL*). The cost of a round-trip trade (*CRT*) is defined as the difference between the per share cost of purchasing and the amount one can sell q (five hundred or five thousand) shares at a certain time ([Domowitz et al., 2005](#); [Griffith and Roseman, 2019](#)). Because *CRT* is essentially a spread measure (if $q = 1$, then *CRT* is equal to the bid-ask spread), we scale it by the quote midpoint and call it $\text{PCRT}_{q,t}$:

$$\begin{aligned} \text{PCRT}_{q,t} &= \frac{\left[\sum_{j=1}^{k-1} S_{j,t} A_{j,t} + \left(q - \sum_{j=1}^{k-1} S_{j,t} \right) A_{k,t} \right] - \left[\sum_{j=1}^{k'-1} D_{j,t} B_{j,t} + \left(q - \sum_{j=1}^{k'-1} D_{j,t} \right) B_{k',t} \right]}{q \times \frac{(A_{1,t} + B_{1,t})}{2}}, \end{aligned} \quad (5)$$

where j indexes bid or ask levels in the order book, k (k') is the last ask (bid) level needed to complete the round-trip transaction, S_j (D_j) is the quoted share depth at ask (bid) level j , and A_j (B_j) is the quoted ask (bid) price at j . We compute $\text{PCRT}_{q,t}$ at each time t when there is a change in the limit order book and then derive the average at each one-minute interval based on how long each $\text{PCRT}_{q,t}$ has prevailed. Finally, we compute the average at the day-level from the one-minute intervals and call it $\text{PCRT}_{q,d}$ for each day d .

We measure for each stock the cumulative price impact of multiple trades, *CumuPriceImpact*, by the slope (λ) of the regression model (see [Goyenko et al., 2009](#); [Hasbrouck, 2009](#)):

$$r_n = \lambda S_n + u_n, \quad (6)$$

where, for the n^{th} five-minute period, r_n is the stock return, S_n is the signed square root dollar volume [i.e.,

$S_n = \sum_k \text{sign}(v_{kn}) \sqrt{|v_{kn}|}$, where v_{kn} is the signed dollar volume of the k th trade in the n th five-minute period], and u_n is the error term.

We use a number of control variables, including closing price of stock i on day d ($\text{Price}_{i,d}$), the Chicago Board Options Exchange (CBOE) Volatility Index on day d (VIX_d), and the market capitalization of stock i on the last day of year $y - 1$ ($\text{MarketCap}_{i,y-1}$). We also use return volatility (TotalVolatility) and the average daily dollar trade volume (DollarVolume) as control variables when appropriate. We measure TotalVolatility by the square root of the sum of squared one-minute quote midpoint returns.

Table 1 shows the mean and standard deviation of each of the above variables (and some other variables to be explained later) for the control and pilot stocks before and after the implementation of the pilot program. We use winsorized values of these variables in the table and in regressions. We winsorize QuotedSpread and PCRT at the 99th percentile values. We winsorize EffectiveSpread , RealizedSpread , TotalVolatility , PricelImpact , CumuPricelImpact , and Autocorrelation at the 1st and 99th percentile values. Table 1 also shows the difference in means between the pre-pilot and pilot periods for each variable, together with the t -statistic. Significant changes in these variables are evident for the control stocks, which could largely reflect changes in the determinants of these variables between the two periods and also possible spillover effects of the pilot program. Changes in the variables for the pilot stocks reflect both the effect of the pilot program and additional changes in these variables that are unrelated to the pilot program. The last column shows the difference-in-differences of means between the control and pilot stocks (pilot – control) for each variable. The results show significant increases in the quoted, effective, and realized spreads, PricelImpact , Depth , CumuDepth , and PAVAIL and significant decreases in CumuPricelImpact and PCRT after the program implementation.⁹ Although useful, the last column results do not accurately measure the effect of the pilot program because they do not control for the effects of other market-wide and firm-specific factors that could also affect the variables.

4. Aggregate effects of the pilot program

Before we test our hypotheses developed in Section 2, we provide evidence regarding the aggregate effect of the tick pilot program on liquidity. We estimate the following regression model using stocks in the pilot and control groups:

$$\text{VAR}_{i,d} = \beta_1 \text{Pilot}_i \times \text{Event}_d + \beta_2 \text{Pilot}_i + \beta_3 \text{Event}_d + \text{ControlVariables}_{i,d} + \lambda_i + \theta_d + \varepsilon_{i,d}, \quad (7)$$

where $\text{VAR}_{i,d}$ represents one of the ten liquidity measures, Pilot_i is equal to one for the pilot (test) stocks and zero

otherwise, Event_d is equal to one after October 2016 (the event date) and zero otherwise, λ_i denotes stock fixed effects, and θ_d denotes day fixed effects. The stock fixed effects capture the effect of time invariant stock attributes on $\text{VAR}_{i,d}$, and the day fixed effects control for the effect of firm-invariant (common or economy-wide) temporal changes in $\text{VAR}_{i,d}$. Following prior research (e.g., Harris, 1994; Chung et al., 2004), we include share price, market capitalization (in log), dollar trading volume (in log), and return volatility as control variables. Following Harris (1994), we use reciprocal of share price in the spread regression models. We also include CBOE Volatility Index (VIX) as a control variable (Rindi and Werner, 2017). We estimate the model with standard errors clustered by both firm and day. To save space, we report the results for the control variables in the Online Appendix (Table OA2 and Table OA3).

4.1. Effects of the pilot program on small and large orders

Panel A of Table 2 shows the results for the quoted spread, effective spread, depth, price impact, and realized spread, and Panel B shows the results for CumuDepth , PAVAIL , PCRT , and CumuPricelImpact . Because the results for the interaction term ($\text{Pilot} \times \text{Event}$) are qualitatively identical with and without stock and day fixed effects, we report only the results without the fixed effects. We report the results with the fixed effects in the Online Appendix (Table OA3). Roughly speaking, the results in Panel A show the effects of the tick pilot program on liquidity measures and market-making revenues that are relevant to small market orders, and those in Panel B show the effects of the program on liquidity measures that are more relevant to large market orders.

The results show that the coefficients on $\text{Pilot} \times \text{Event}$ are positive and significant in the quoted and effective spread regressions, indicating that the pilot stocks have higher quoted and effective spreads compared with the control stocks after the event date (i.e., in the pilot period). Both quoted and effective spreads increase, on average, by 15 basis points. The coefficient on $\text{Pilot} \times \text{Event}$ in the depth regression is positive and significant, indicating that liquidity providers quote larger depths in the pilot stocks compared with the control stocks after the event. We find an increase in realized spreads, despite a concurrent increase in price impact, indicating an increase in market-making revenues, net of losses to informed traders. The coefficients on Pilot are not significant in all regressions in Panel A, indicating that the pilot and control stocks are similar in these metrics during the pre-pilot period, which is expected given that both the control and pilot stocks are randomly selected from the same pool of stocks.

Panel B shows that the coefficients on $\text{Pilot} \times \text{Event}$ in the regression models of CumuDepth and PAVAIL are all positive and significant and that the corresponding coefficients in the regression models of PCRT (i.e., the cost of a round-trip trade of five hundred and five thousand shares) are all negative and significant, indicating that the tick pilot program increases available liquidity for large orders in the pilot stocks. These results could be attributed, at least in part, to the increase in the NBBO depth shown in

⁹ We report the results using the realized spreads (one minute), PricelImpact (one minute), CumuDepth (5 percent), and PAVAIL (5 percent). We obtain similar results when we use the realized spread (five minutes), PricelImpact (five minutes), CumuDepth (20 cents), and PAVAIL (20 cents) in the regression. We provide these results in the Online Appendix.

Table 1

Descriptive statistics.

This table shows the mean and standard deviation of the variables for the control and pilot stocks before and after the implementation of the pilot program. The table also shows the difference in means between the pre-pilot and pilot periods for each variable, together with the *t*-statistic. We define October 1, 2015 to September 23, 2016 as the pre-pilot period and October 31, 2016 to October 31, 2017 as the pilot period. The last column shows the result of the difference-in-differences test for each variable. *QuotedSpread* is the time-weighted average percentage quote spread. *EffectiveSpread* is the share-weighted average percentage effective spread. *PricelImpact* is the share-weighted average quote midpoint return. *RealizedSpread* is the share-weighted average percentage realized spread. *TBC* is the percentage of time during which quoted spreads are equal to or smaller than 5 cents in the pre-pilot period. *Depth* is the time-weighted average quoted depth at the national best bid and offer (NBBO). *CumuDepth* is the time-weighted cumulative depth available on both sides of the limit order book within 5% of the NBBO. *PAVAIL* is the percentage of one-minute intervals during the trading day when five thousand shares are available on both sides of the limit order book within 5% of the NBBO. *PCRT* is the average per share cost of a round-trip trade of certain number of share scaled by the quote midpoint. *CumuPricelImpact* is the cumulative price impact of multiple trades measured by the slope of regression of five minute return on five-minute signed square root of dollar volume. *TotalVolatility* is the square root of the sum of squared one-minute quote midpoint returns. *FundamentalVolatility* is the difference between *TotalVolatility* and the standard deviation of the pricing error. We measure firm-specific information by $1 - R^2$, where R^2 is obtained by regressing daily stock excess return on [Fama and French \(1992\)](#) three factors (i.e., equity premium, return difference between small and large stocks, and return difference between high and low book-to-market stocks) for each quarter. *Autocorrelation* is the R^2 from the regression of $Return_t$ on $Return_{t-1}$, where $Return_t$ is the quote midpoint return from minute $t - 1$ to minute t . *OIB_Predictability* is the R^2 from regressing one-minute stock return on the order imbalance in the previous one minute-interval, where order imbalance is the difference in dollar trading volume between buyer- and seller-initiated trades. *DollarVolume* is the average daily dollar trade volume. *Trades* is the daily total number of trades. *TradeSize* is the average dollar trade size. *Price* is the closing price. *MarketCap* is market capitalization (shares outstanding $\times$ share price). Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Variable	Control group					Pilot group					(B) – (A)
	Pre-pilot period		Pilot period		Difference in means between the pre-pilot and pilot periods (A)	Pre-pilot period		Pilot period		Difference in means between the pre-pilot and pilot periods (B)	
	Mean	Standard deviation	Mean	Standard deviation		Mean	Standard deviation	Mean	Standard deviation		
Liquidity variables											
<i>QuotedSpread</i>	0.6268	0.8162	0.5662	0.7283	–0.0606*** (–25.08)	0.6019	0.7794	0.7118	0.7495	0.1099*** (45.50)	0.1705*** (49.97)
<i>EffectiveSpread</i>	0.2671	0.3400	0.2339	0.3015	–0.0333*** (–33.17)	0.2568	0.3234	0.3759	0.3821	0.1190*** (106.55)	0.1523*** (101.51)
<i>RealizedSpread</i>	0.1446	0.2944	0.1289	0.2722	–0.0156*** (–17.67)	0.1398	0.2829	0.1952	0.3244	0.0555*** (57.67)	0.0711*** (54.42)
<i>TBC</i>	0.5665	0.3259				0.5567	0.3315				
<i>PricelImpact</i>	0.1226	0.1962	0.1049	0.1767	–0.0177*** (–30.28)	0.1171	0.1853	0.1805	0.2304	0.0634*** (96.07)	0.0811*** (92.10)
<i>Depth</i>	18.166	25.076	26.251	144.13	8.0851*** (25.25)	19.080	18.532	96.799	483.14	77.719*** (72.51)	69.634*** (61.96)
<i>CumuDepth</i>	4217.4	4888.5	4681.9	28,960	464.53*** (7.08)	4674.5	7273.6	11,158	27,872	6483.2*** (100.16)	6018.7*** (64.45)
<i>PAVAIL</i>	0.0377	0.1596	0.0341	0.1591	–0.0036*** (–7.17)	0.0530	0.1964	0.1721	0.3461	0.1192*** (132.95)	0.1228*** (118.94)
<i>PCRT500</i>	3.6662	3.8579	3.5675	3.6934	–0.0987*** (–8.22)	3.4620	3.5886	2.7019	2.9484	–0.7601*** (–72.29)	–0.6614*** (–41.45)
<i>PCRT5000</i>	19.968	13.654	21.273	14.672	1.3052*** (28.97)	19.375	13.719	17.323	13.509	–2.0520*** (–47.13)	–3.3571*** (–53.55)
<i>CumuPricelImpact</i>	6.7663	13.644	5.9848	11.831	–0.7815*** (–19.58)	6.4535	13.184	4.9041	10.855	–1.5494*** (–40.52)	–0.7679*** (–13.92)
Information variables											
<i>Fundamental-Volatility</i>	0.0220	0.0155	0.0191	0.0127	–0.0029*** (–59.25)	0.0210	0.0146	0.0186	0.0120	–0.0024*** (–51.34)	0.0006*** (9.00)
$1 - R^2$	72.824	17.453	75.378	18.012	2.5539*** (5.98)	72.352	17.523	75.855	17.679	3.5029*** (8.16)	0.9490 (1.57)
<i>Autocorrelation</i>	2.6415	6.0360	2.6842	6.3393	0.0427** (2.2)	2.6319	5.9786	2.6584	6.4844	0.0266 (1.33)	–0.0161 (–0.58)
<i>OIB_Predictability</i>	5.1636	14.026	4.7544	13.167	–0.4092*** (–9.22)	5.1266	13.919	4.2852	12.477	–0.8415*** (–19.08)	–0.4323*** (–6.93)
Trading activities											
<i>DollarVolume</i>	4.9881	9.9526	6.4419	18.582	1.4538*** (31.48)	5.3260	10.786	6.3367	13.243	1.0107*** (26.58)	–0.4430*** (–7.34)
<i>Trades</i>	1759.2	2334.0	2000.8	3158.8	241.58*** (28.02)	1843.7	2464.2	1811.4	2456.2	–32.334*** (–4.17)	–273.91*** (–23.51)
<i>TradeSize</i>	2302.2	2072.8	2489.5	2059.3	187.30*** (29.07)	2425.6	2337.3	2774.6	2447.0	348.99*** (46.19)	161.69*** (16.31)
Control variables											
<i>TotalVolatility</i>	0.0242	0.0165	0.0205	0.0135	–0.0037*** (–78.27)	0.0231	0.0155	0.0200	0.0126	–0.0031*** (–68.94)	0.0007*** (11.38)
<i>Price</i>	23.907	26.914	28.882	33.272	4.9753*** (53.49)	24.953	30.898	29.862	37.504	4.9090*** (45.86)	–0.0663 (–0.47)
<i>MarketCap</i>	718.87	724.09	795.09	802.63	76.219*** (32.28)	753.09	724.53	833.56	805.04	80.469*** (33.59)	4.2503 (1.26)

Table 2

Effects of the pilot program on various measures of liquidity.

This table shows the results of the following regression model: $VAR_{i,d} = \beta_1 Pilot_i \times Event_d + \beta_2 Pilot_i + \beta_3 Event_d + Control\ Variables_{i,d} + \varepsilon_{i,d}$, where $VAR_{i,d}$ represents one of the ten liquidity measures, $Pilot_i$ is equal to one for the pilot stocks and zero otherwise, and $Event_d$ is equal to one after October 2016 (the event date) and zero otherwise. We include Chicago Board Options Exchange Volatility Index (VIX), share price, market capitalization (in log), dollar trading volume (in log), and return volatility as control variables. We report the results for the control variables in the Online Appendix (Table OA.2). Figures in parentheses are *t*-statistics. Standard errors are clustered by both firm and day. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *N* denotes the number of observations. See Table 1 for the definition of each dependent variable. NBBO is national best bid and offer.

Panel A: Regression results					
Variable	QuotedSpread	EffectiveSpread	Depth	PricelImpact	RealizedSpread
<i>Pilot</i> × <i>Event</i>	0.1513*** (10.26)	0.1514*** (21.48)	70.951*** (8.81)	0.0790** (27.25)	0.0729*** (14.57)
<i>Pilot</i>	0.0099 (0.54)	0.0028 (0.46)	−0.4921 (−0.47)	0.0020 (1.00)	0.0008 (0.17)
<i>Event</i>	0.0518*** (6.28)	0.0049 (1.62)	0.8307 (0.26)	−0.0006 (−0.46)	0.0055** (2.02)
<i>N</i>	805,555	805,646	805,555	804,448	803,954
Adjusted <i>R</i> ²	0.5537	0.4769	0.0248	0.2510	0.2783
<i>F</i> -statistic	353.81	457.97	95.67	566.86	340.89
Panel B: Regression results					
Variable	CumuDepth	PAVAIL	PCRT500	PCRT5000	CumuPricelImpact
<i>Pilot</i> × <i>Event</i>	5.0770*** (6.26)	0.1163*** (12.62)	−0.7353*** (−12.50)	−3.6121*** (−13.92)	−1.0608*** (−8.43)
<i>Pilot</i>	0.4794* (1.76)	0.0168** (2.17)	−0.0752 (−0.70)	−0.3811 (−0.69)	0.1597 (0.97)
<i>Event</i>	0.2865 (0.48)	−0.0171*** (−4.24)	0.1909*** (4.23)	2.0782*** (12.07)	0.5583*** (5.91)
<i>N</i>	777,589	777,589	777,589	777,589	804,200
Adjusted <i>R</i> ²	0.0330	0.1132	0.4424	0.1959	0.2922
<i>F</i> -statistic	19.65	29.88	192.87	102.15	326.22
Panel C: Replication of Panel B after omitting NBBO quotes					
Variable	CumuDepth _{w/oNBBO}	PAVAIL _{w/oNBBO}	PCRT500 _{w/oNBBO}	PCRT5000 _{w/oNBBO}	
<i>Pilot</i> × <i>Event</i>	3.6721*** (5.53)	0.0915*** (10.77)	0.5430*** (4.19)	−2.1117*** (−7.79)	
<i>Pilot</i>	0.4236* (1.70)	0.0157** (2.27)	−0.1103 (−0.56)	−0.4043 (−0.67)	
<i>Event</i>	0.1450 (0.30)	−0.0129*** (−3.75)	0.4987*** (5.04)	2.2975*** (12.52)	
<i>N</i>	777,589	777,589	777,588	777,588	
Adjusted <i>R</i> ²	0.0328	0.0889	0.4390	0.2102	
<i>F</i> -statistic	17.42	21.63	136.74	95.10	

Panel A. The coefficient on *Pilot* × *Event* in the regression model of *CumuPricelImpact* is negative and significant, suggesting that the pilot program decreases the price impact of large orders. Overall, these results indicate that the tick pilot program increases liquidity for large orders.

4.2. Effects of the pilot program on the control stocks

Rindi and Werner (2017) show that the pilot program has a spillover effect; i.e., it affects not only the stocks in the treatment group (stocks in the three test groups) but also the stocks in the control group. Rindi and Werner (2017) analyze the spillover effect using only the NBBO quotes, which capture largely the spillover effect for small orders. Our results in Table 2 show the spillover effect for both large and small orders. As to the spillover effect for smaller orders, Panel A shows that the coefficients on

Event are significant and positive in the regression models of the quoted and realized spreads, suggesting that the pilot program has spillover effects on the control stocks. The increase in quoted and realized spreads for the control stocks is similar to the finding of Rindi and Werner (2017). The negative (positive) and significant coefficients on *Event* in the regression model of *PAVAIL* (*PCRT* and *CumuPricelImpact*) in Panel B suggest that the pilot program also reduces available liquidity for large orders in the control stocks. As noted in Rindi and Werner (2017), these results could be driven by market-wide changes in liquidity provision. To rule out the effects of market-wide changes, we focus on fixed effect panel regressions in later parts of the paper. We measure cross-sectional variations in spillover effects using the interaction term of *EVENT* and *TBC*. Whether the above results are driven entirely by the effect of the pilot program on the top of the book is unclear.

4.3. Effects of the pilot program beyond the top of the book

This section explores whether the effect of the tick pilot program on liquidity is limited only to the top of the book for the treatment and control stocks. For this, we reconstruct liquidity measures (except *CumuPriceImpact*) in Panel B of Table 2 after dropping NBBO quotes, reestimate the regression models, and provide the results in Panel C. The results show that the coefficients on *Pilot* $\times$ *Event* in the regression models of *CumuDepth*_{w/oNBBO} and *PAVAIL*_{w/oNBBO} are positive and significant, suggesting that the positive effect of the pilot program on liquidity extends beyond the top of the book.

We find mixed results for the effect of the pilot program on *PCRT*. As in Panel B, we find that the coefficient on *PCRT5000*_{w/oNBBO} is negative and significant, indicating that the pilot program reduces the expected execution cost of large orders even if we ignore its effect on the top of the book liquidity. The coefficient on *PCRT500*_{w/oNBBO} is positive and significant, indicating that the pilot program increases the expected execution cost of relatively small orders if we ignore its effect on the top of the book liquidity. These results suggest that the decrease in the cost of a round-trip trade of five hundred shares shown in Panel B can be largely attributed to the effect of the pilot program on the top of the book liquidity.

As in Panel B, we find negative (positive) and significant coefficients on *Event* in the regression model of *PAVAIL*_{w/oNBBO} (*PCRT*_{w/oNBBO}), suggesting that the pilot program reduces available liquidity for large orders in the control stocks even if we ignore its effect on the top of the book liquidity. We interpret this result as evidence that the spillover effects extend beyond the top of the book.

5. Effects of the quote, trade, and trade-at rules on liquidity

In this section, we conduct empirical tests of Hypothesis 1 through Hypothesis 4 to shed some light on underlying mechanisms and channels behind the results provided in Section 4 by analyzing the effects of the quote (*Q*), trade (*T*), and trade-at (*TA*) rules separately. We use the following regression model to analyze these effects:

$$\begin{aligned} VAR_{i,d} = & \beta_1 Q_i \times Event_d \times TBC_{i,d} + \beta_2 Q_i \times Event_d \\ & + \beta_3 Event_d \times TBC_{i,d} + \beta_4 T_i \times Event_d + \beta_5 TA_i \\ & \times Event_d + Control Variables_{i,d} + \lambda_i + \theta_d + \varepsilon_{i,d}, \end{aligned} \quad (8)$$

where $VAR_{i,d}$ represents one of the liquidity measures, Q_i , T_i , and TA_i are the quote, trade, and trade-at rule dummy variables, $Event_d$ is the tick pilot dummy variable (which is equal to one for the tick pilot period and zero otherwise), $TBC_{i,d}$ is the extent to which the new quote tick size (\$0.05) is a binding constraint on quoted spreads in the pilot period, λ_i denotes stock fixed effects, and θ_d denotes day fixed effects.

Following prior research, we again include the CBOE Volatility Index (*VIX*), share price, market capitalization (in log), dollar trading volume (in log), and return volatility as control variables. We estimate the model with standard

errors clustered by both firm and day. We obtain qualitatively similar results with and without the fixed effects. For brevity, we report only the results with the fixed effects. We provide the results without the fixed effects and the results for the control variables in the Online Appendix (Table OA4 and Table OA5).

5.1. Results for the quoted and effective spreads (Hypothesis 1)

The first column in Table 3 shows the results for the quoted spread. The coefficient on $Q \times Event$ (−0.0950) is negative and significant, indicating that the quoted spread during the pilot period is smaller than the corresponding value during the pre-pilot period for stocks with spreads that are always greater than 5 cents during the pre-pilot period ($TBC = 0$). These results suggest that the larger quote tick size (the *Q* rule) results in a decrease in quoted spreads for stocks with large spreads during the pre-pilot period, which is consistent with the finding of Rindi and Werner (2017) that quoted spreads decrease for unconstrained stocks (i.e., stocks with large spreads during the pre-pilot period).

The coefficient on $Q \times Event \times TBC$ is positive and significant, indicating that among stocks with non zero *TBC*, the *Q* rule results in a greater increase in quoted spreads for stocks with larger *TBC* values. These results are consistent with Hypothesis 1 that the *Q* rule results in a greater increase in quoted spreads for stocks with larger *TBC* because the new quote tick size is more frequently a binding constraint on quoted spreads in the pilot period. For stocks with $TBC = 1$, the effect of the *Q* rule on quoted spreads can be measured by adding the coefficient on $Q \times Event \times TBC$ and the coefficient on $Q \times Event$, which is equal to $0.3577 (= 0.4527 - 0.0950)$. More generally, the *Q* rule results in an increase in quoted spreads for stocks with $[0.4527 \times (Q \times Event \times TBC)] - [0.0950 \times (Q \times Event)] < 0$ or, alternatively, $TBC > 0.2099$. That is, the *Q* rule results in an increase in quoted spreads for stocks with *TBC* values larger than 0.2099, which corresponds to the 25th percentile value of *TBC* for the pilot stocks, indicating that the *Q* rule results in an increase in quoted spreads for 75% of the pilot stocks.

Overall, these results show that whether the *Q* rule increases or decreases quoted spreads depends on the extent to which the new (large) quote tick size is a binding constraint on quoted spreads. The majority of stocks in the pilot group experienced an increase in quoted spreads after the quote tick size increase, which is generally consistent with the prediction of the theoretical models of Glosten (1994), Biais et al. (2000), and Sandås (2001). These models suggest that liquidity suppliers provide liquidity at each price level until their marginal expected profits of providing another share are zero and predict that quoted spreads increase with an increase in the tick size. Comerton-Forde et al. (2019) find that the pilot program results in an increase in dark market share for stocks in TG1 and TG2 and interpret the result as evidence that the increase in the tick size moves traders from lit markets to dark venues for better execution prices within the spread (Kwan et al., 2015). Buti et al. (2015) predict that dark trading has a negative

Table 3

Effects of the quote (*Q*), trade (*T*), and trade-at (*TA*) rules on spreads, depths, and price impact.

This table shows the results of the following regression model: $VAR_{i,d} = \beta_1 Q_i \times Event_d \times TBC_{i,d} + \beta_2 Q_i \times Event_d + \beta_3 Event_d \times TBC_{i,d} + \beta_4 T_i \times Event_d + \beta_5 TA_i \times Event_d + Control\ Variables_{i,d} + \lambda_i + \theta_d + \varepsilon_{i,d}$, where $VAR_{i,d}$ represents *QuotedSpread*, *EffectiveSpread*, *Depth*, *PricelImpact*, or *RealizedSpread*. Q_i is the quote rule dummy variable (which is equal to one for stocks that are quoted in \$0.05 increments and zero otherwise), T_i is the trade rule dummy variable (which is equal to one for stocks that are traded in \$0.05 increments and zero otherwise), TA_i is the trade-at rule dummy variable (which is equal to one for stocks that are subject to the trade-at requirement and zero otherwise), $Event_d$ is the tick pilot dummy variable (which is equal to one for the tick pilot period and zero otherwise), $TBC_{i,d}$ is the extent to which the new quote tick size (\$0.05) is a binding constraint on quoted spreads in the pilot period (measured by the percentage of time during which quoted spreads are equal to or smaller than 5 cents in the pre-pilot period), λ_i denotes stock fixed effects, and θ_d denotes day fixed effects. We include share price, market capitalization (in log), dollar trading volume (in log), and return volatility as control variables. We report the results for the control variables in the Online Appendix (Table OA.4 and Table OA.5). Figures in parentheses are *t*-statistics. Standard errors are clustered by both firm and day. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *N* denotes the number of observations. See Table 1 for the definition of each dependent variable.

Variable	QuotedSpread	EffectiveSpread	Depth	PricelImpact	RealizedSpread
$Q \times Event \times TBC$	0.4527*** (9.90)	0.2224*** (11.16)	138.06*** (4.58)	0.0898*** (11.69)	0.1332*** (8.93)
$Q \times Event$	-0.0950*** (-2.74)	0.0225* (1.67)	-20.718 (-1.37)	0.0202*** (3.73)	0.0025 (0.24)
$Event \times TBC$	0.1456*** (4.88)	0.0717*** (7.09)	15.865 (1.33)	0.0299*** (7.07)	0.0418*** (5.22)
$T \times Event$	0.0186 (0.73)	0.0283** (2.07)	20.487 (1.14)	0.0113** (2.23)	0.0172* (1.71)
$TA \times Event$	-0.0251 (-0.97)	-0.0357** (-2.55)	9.9745 (0.49)	0.0054 (0.93)	-0.0415*** (-4.24)
Control variables	Yes	Yes	Yes	Yes	Yes
Stock and day fixed effects	Yes	Yes	Yes	Yes	Yes
<i>N</i>	805,555	805,646	805,555	803,954	803,954
Adjusted R^2	0.7571	0.6069	0.1310	0.2907	0.3824
<i>F</i> -statistic	249.81	244.56	55.87	375.34	112.18

effect on the liquidity of regular exchanges with less liquid limit order books.¹⁰ Hence, the decrease (increase) in displayed liquidity (quoted spreads) could also be, at least in part, due to the migration of liquidity from lit markets to dark pools.

The coefficients on $T \times Event$ and $TA \times Event$ are insignificant, indicating that both the *T* and *TA* rules do not affect quoted spreads. The insignificant effect of the *TA* rule on quoted spreads does not necessarily refute implication 15 that the *TA* rule increases visible liquidity, as long as the *TA* rule increases liquidity at any position in the limit order book.

The second column in Table 3 shows the results for the effective spread. The coefficient on $Q \times Event$ is positive and significant, indicating that the *Q* rule results in an increase in effective spreads even for stocks with large spreads during the pre-pilot period ($TBC = 0$). Not surprisingly and consistent with the results for the quoted spread, the coefficient on $Q \times Event \times TBC$ is positive and significant, indicating that the *Q* rule results in a greater increase in effective spreads for stocks with larger *TBC*, supporting Hypothesis 1. In contrast to the results for the quoted spread, the coefficient on $T \times Event$ is positive and significant, which supports Hypothesis 1 that a larger trade tick size increases the effective spread because liquidity suppliers are less likely to provide price improvements when the trade tick size is larger. The coefficient on $TA \times Event$ is negative and significant, which also supports Hypothesis 1

that the trade-at rule reduces the effective spread by increasing displayed liquidity. Displayed liquidity is not limited to liquidity at the NBBO. Any increase in liquidity in the limit order book could reduce the effective spread of a marker order that is larger than the depth at the NBBO. Hence, even if the *TA* rule does not have an effect on the quoted spread, it can still have an effect on the effective spread.

5.2. Results for price impact and the realized spread (Hypothesis 2)

The fourth column in Table 3 shows the results for price impact. The coefficients on $Q \times Event$ and $Q \times Event \times TBC$ are positive and significant, indicating that the *Q* rule increases price impact and this increase is greater for stocks with larger *TBC* values, supporting Hypothesis 3. The coefficient on $T \times Event$ is positive and significant, suggesting that the trade rule increases the price impact of a trade. This result is consistent with the idea that traders have greater incentives to gather and trade on private information when the trade tick size is larger.

Because the *Q* rule increases both the effective spread and price impact, especially for stocks with large *TBC* values, we expect the *Q* rule to increase (decrease) the realized spread if its effect on the effective spread is greater (smaller) than its effect on price impact (given that the realized spread = the effective spread – price impact). The last column in Table 3 shows the results for the realized spread. We find that although the coefficient on $Q \times Event$ is not significantly different from zero, the coefficient on

¹⁰ Pilot stocks are considered small and illiquid.

$Q \times \text{Event} \times \text{TBC}$ is positive and significant. These results indicate that the Q rule increases the realized spread and this increase is larger when the new tick size is more frequently a binding constraint on spreads in the pilot period, suggesting that its effect on the effective spread is greater than its effect on price impact. This result is consistent with the finding of Comerton-Forde et al. (2019).

Because the T rule increases both the effective spread and price impact, we expect the T rule to increase (decrease) the realized spread if its effect on the effective spread is greater (smaller) than its effect on price impact. The coefficient on $T \times \text{Event}$ is positive and significant, indicating that the effect of the T rule on the effective spread is greater than its effect on price impact. Consistent with Hypothesis 3, the trade-at rule significantly reduces the realized spread (i.e., the coefficients on $TA \times \text{Event}$ are all negative and significant). This result can also suggest that the trade-at rule reduces profits for liquidity providers, perhaps by increasing competition for small trades by moving previously undisplayed liquidity onto lit exchanges.

5.3. Results for the NBBO and cumulative depths and PAVAIL (Hypothesis 3)

The third column in Table 3 shows the results for the depth at the NBBO. The coefficient on $Q \times \text{Event}$ is not significantly different from zero, indicating that the quote rule does not have an effect on the inside depth for stocks with zero TBC . The coefficient on $Q \times \text{Event} \times \text{TBC}$ is positive and significant, indicating that, among stocks with positive TBC , the Q rule results in a greater increase in quoted depths for stocks with larger TBC . These results are consistent with Hypothesis 3 that the larger quote tick size results in an increase in quoted depths for stocks with positive TBC because the new tick size is more frequently a binding constraint on quoted spreads in the pilot period. Although the coefficient on $T \times \text{Event}$ is positive, it is not significant, indicating either that the trade tick size does not affect front running or that front running has no impact on liquidity provision. Overall, these results suggest that the larger tick size affects the NBBO depth mainly through the binding constraint on spreads.

The first two columns in Table 4 show the effects of the three rules on the cumulative depth within 5% of the NBBO and PAVAIL. We find that the coefficients on $Q \times \text{Event}$ are negative and significant, but the coefficients on $Q \times \text{Event} \times \text{TBC}$ are positive and significant in both regressions, which again suggests that the effect of the Q rule depends on TBC . We find similar results using the total number of shares available in the limit order book. These results are generally consistent with Hypothesis 3 that the Q rule increases the cumulative depth and PAVAIL, especially for those stocks with large TBC values. Although the coefficients on $T \times \text{Event}$ are positive, they are not significant in both regressions, indicating either that the trade tick size does not affect front running or that front running has no impact on liquidity provision. Likewise, the coefficient on $TA \times \text{Event}$ is not significant, indicating the TA rule does not materially affect the cumulative depth and PAVAIL. Overall, these results are similar to the results for

the NBBO depth that the larger tick size affects market depths mainly through the binding constraint on spreads.

5.4. Results for the cost of a round-trip trade (PCRT) and the cumulative price impact (Hypothesis 4)

The third and fourth columns in Table 4 show the results for $PCRT500$ and $PCRT5000$ (i.e., the cost of a round-trip trade of five hundred and five thousand shares). The coefficient on $Q \times \text{Event}$ is negative and significant in the $PCRT500$ regression, indicating that the Q rule reduces the round-trip trading cost of five hundred shares for stocks with large pre-pilot period spreads (stocks with zero TBC). The coefficient on $Q \times \text{Event} \times \text{TBC}$ is insignificant, suggesting that the cost of trading five hundred shares does not change with TBC . The coefficient on $Q \times \text{Event} \times \text{TBC}$ is negative and significant in the regression model of $PCRT5000$, indicating that the round-trip trading cost of five thousand shares is smaller for stocks with larger TBC values. These results are opposite of Griffith and Roseman (2019), who find that for constrained stocks trading costs increase for large trades (see their Panel A of Table 5). One possible explanation for the different findings is that Griffith and Roseman (2019) use only limit orders on Nasdaq, whereas we use available liquidity across all US trading venues. Griffith and Roseman (2019) use only Nasdaq-listed stocks, whereas we use Nasdaq- and NYSE-listed stocks. We show that our results are robust to using only Nasdaq-listed stocks (see Table OA8 in the Online Appendix). Griffith and Roseman (2019) measure transaction costs in dollars. We find that our results are robust when we replicate our analysis using transaction costs measured in dollars (see Table OA9 in the Online Appendix). Overall, our results are consistent with Hypothesis 4 that the Q rule reduces the execution cost of large orders and the reduction is larger for stocks with higher TBC , perhaps because of a larger increase in depths in the limit order book.

The coefficients on $T \times \text{Event}$ are not significant in all three regressions, indicating either that the trade tick size does not affect front running or that front running has no impact on liquidity provision. We obtain mixed results for the coefficients on $TA \times \text{Event}$ (i.e., one is negative and significant, while the other two are insignificant), suggesting that the TA rule reduces only the execution cost of the smaller order (five hundred shares). Overall, these results are similar to the results for the NBBO depth, cumulative depth, and PAVAIL that the larger tick size affects liquidity mainly through the binding constraint on spreads.

We show earlier that the effective spread increased after the pilot program implementation. The effective spread is estimated using all trades from both small and large orders, whereas $PCRT$ is the estimate of trading cost for a given order size. Hence, $PCRT$ is a better metric than the effective spread to assess the effect of the pilot program on the execution cost of large orders. Our results suggest that traders would have incurred smaller execution costs trading the pilot stocks than the control stocks in the pilot period had they submitted large orders. There are other reasons that effective spreads and $PCRT$ are not directly comparable and perhaps do not change in the same direction. We calculate $PCRT$ whenever a change occurs in the

Table 4

Effects of the quote (Q), trade (T), and trade-at (TA) rules on *CumuDepth*, *PAVAIL*, *PCRT500*, *PCRT5000*, or *CumuPriceImpact*.

This table shows the results of the following regression model: $VAR_{i,d} = \beta_1 Q_i \times Event_d \times TBC_{i,d} + \beta_2 Q_i \times Event_d + \beta_3 Event_d \times TBC_{i,d} + \beta_4 T_i \times Event_d + \beta_5 TA_i \times Event_d + Control\ Variables_{i,d} + \lambda_i + \theta_d + \varepsilon_{i,d}$, where $VAR_{i,d}$ represents *CumuDepth*, *PAVAIL*, *PCRT500*, *PCRT5000*, or *CumuPriceImpact*. Q_i is the quote rule dummy variable (which is equal to one for stocks that are quoted in \$0.05 increments and zero otherwise), T_i is the trade rule dummy variable (which is equal to one for stocks that are traded in \$0.05 increments and zero otherwise), TA_i is the trade-at rule dummy variable (which is equal to one for stocks that are subject to the trade-at requirement and zero otherwise), $Event_d$ is the tick pilot dummy variable (which is equal to one for the tick pilot period and zero otherwise), $TBC_{i,d}$ is the extent to which the new quote tick size (\$0.05) is a binding constraint on quoted spreads in the pilot period (measured by the percentage of time during which quoted spreads are equal to or smaller than 5 cents in the pre-pilot period), λ_i denotes stock fixed effects, and θ_d denotes day fixed effects. We include share price, market capitalization (in log), dollar trading volume (in log), and return volatility as control variables. We report the results for the control variables in the Online Appendix (Table OA.6 and Table OA.7). Figures in parentheses are *t*-statistics. Standard errors are clustered by both firm and day. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *N* denotes the number of observations. See Table 1 for the definition of each dependent variable.

Variable	CumuDepth	PAVAIL	PCRT500	PCRT5000	CumuPriceImpact
$Q \times Event \times TBC$	13.483*** (4.24)	0.3302*** (11.79)	0.2683 (1.54)	-4.2035*** (-6.05)	-2.5224*** (-7.23)
$Q \times Event$	-3.2960** (-2.38)	-0.0765*** (-5.63)	-0.7694*** (-5.47)	-0.7311 (-1.63)	0.2400 (0.94)
$Event \times TBC$	1.7408 (0.74)	-0.0314** (-2.19)	0.5732*** (4.49)	-0.0607 (-0.14)	1.5281*** (6.00)
$T \times Event$	0.7700 (0.76)	0.0022 (0.13)	0.0544 (0.58)	-0.3278 (-0.72)	0.0187 (0.09)
$TA \times Event$	1.6827 (1.05)	0.0186 (0.99)	-0.2082** (-2.17)	-0.3051 (-0.64)	0.3811* (1.75)
Control variables	Yes	Yes	Yes	Yes	Yes
Stock and day fixed effects	Yes	Yes	Yes	Yes	Yes
<i>N</i>	777,589	777,589	777,589	777,589	804,200
Adjusted R^2	0.3560	0.6715	0.8289	0.8511	0.3549
<i>F</i> -statistic	25.98	26.71	154.95	55.67	248.42

Table 5

Effects of the quote (Q), trade (T), and trade-at (TA) rules on Rule 605 effective spreads.

This table shows the results of the following regression model: $EffectiveSpread_{i,m} = \beta_1 Q_i \times Event_m \times TBC_{i,m} + \beta_2 Q_i \times Event_m + \beta_3 Event_m \times TBC_{i,m} + \beta_4 T_i \times Event_m + \beta_5 TA_i \times Event_m + Control\ Variables_{i,m} + \lambda_i + \theta_m + \varepsilon_{i,m}$, where $EffectiveSpread_{i,m}$ is the average percentage effective spread of stock *i* in month *m* from Rule 605 reports. Q_i is the quote rule dummy variable (which is equal to one for stocks that are quoted in \$0.05 increments and zero otherwise), T_i is the trade rule dummy variable (which is equal to one for stocks that are traded in \$0.05 increments and zero otherwise), TA_i is the trade-at rule dummy variable (which is equal to one for stocks that are subject to the trade-at requirement and zero otherwise), $Event_m$ is the tick pilot dummy variable (which is equal to one for the tick pilot period and zero otherwise), $TBC_{i,m}$ is the extent to which the new quote tick size (\$0.05) is a binding constraint on quoted spreads in the pilot period (measured by the percentage of time during which quoted spreads are equal to or smaller than 5 cents in the pre-pilot period), λ_i denotes stock fixed effects, and θ_m denotes month fixed effects. We include share price, market capitalization (in log), dollar trading volume (in log), and return volatility as control variables. We report the results for the control variables in the Online Appendix (Table OA.10). Figures in parentheses are *t*-statistics. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *N* denotes the number of observations.

Variable	Order size group			
	100–499	500–1999	2000–4999	5000+
$Q \times Event \times TBC$	0.0822** (2.49)	0.0046 (0.08)	-0.4390*** (-2.94)	-1.2298*** (-3.02)
$Q \times Event$	0.0132 (0.53)	-0.0354 (-0.81)	0.2394* (1.86)	0.8975** (2.43)
$Event \times TBC$	0.1874*** (7.97)	0.1448*** (3.69)	0.4652*** (4.32)	0.8981*** (3.05)
$T \times Event$	0.0040 (0.23)	-0.0143 (-0.54)	0.0078 (0.12)	0.2824* (1.95)
$TA \times Event$	-0.0658*** (-3.61)	0.0028 (0.10)	-0.0015 (-0.02)	0.0213 (0.15)
Control variables	Yes	Yes	Yes	Yes
Stock and month fixed effects	Yes	Yes	Yes	Yes
<i>N</i>	31,257	19,773	7175	2264
Adjusted R^2	0.4219	0.3113	0.2340	0.2802
<i>F</i> -statistic	133.83	29.32	10.32	3.00

limit order book throughout the day. We calculate effective spreads only when trade occurs, which is endogenously determined. Patient traders can submit large orders only when liquidity is high. In addition, the pilot program could have moved hidden liquidity in the pilot stocks to exchanges because the trade rule (i.e., larger trade tick size) decreases front running by increasing the cost of front running and the trade-at rule gives liquidity suppliers the incentive to display instead of hide their trading interests. As a result, effective spreads increased while *PCRT* decreased relative to the pre-pilot program and the control stocks. Overall, these results suggest that the larger quote tick size benefits large institutional traders.

The last column in Table 4 shows the results for the cumulative price impact (*CumuPriceImpact*), which captures the execution cost of traders who split their large orders into smaller ones. We find that while the coefficient on $Q \times Event$ is not significant, the coefficient on $Q \times Event \times TBC$ is negative and significant. These results are generally consistent with Hypothesis 4 that the *Q* rule decreases the cumulative price impact, especially for those stocks with high *TBC* values. Another possible reason that the *Q* rule reduces the cumulative price impact is that liquidity providers are less responsive to adverse selection risks (and costs) because the larger quote tick size allows them to post larger than equilibrium spreads. The coefficient on $T \times Event$ is insignificant, suggesting either that the trade tick size does not affect front running or that front running has no impact on liquidity providers. The coefficient on $TA \times Event$ is positive and significant at the 10% level, suggesting that the cumulative price impact increases for stocks subject to the trade-at rule.

5.5. Effects of the pilot program on Rule 605 effective spreads

To further assess the effect of the pilot program on liquidity for small and large orders, we estimate regression model Eq. (8) using Rule 605 data. The SEC Rule 605 requires market centers to report the quality of executions. These reports include information at the stock-month level about average effective spreads from market orders for each of the following four order size groups: 100–499 shares, 500–1999 shares, 2000–4999 shares, and 5000 or more shares. We obtain Rule 605 reports for our sample from the exchange websites and the Transaction Auditing Group and define October 2015 to September 2016 as the pre-pilot period and November 2016 to October 2017 as the pilot period. All observations of the Investors Exchange (IEX) are dropped due to its transition from an alternative trading system (ATS) to a national securities exchange in September 2016. All observations of the NYSE National (NSX) are dropped because it was acquired by the NYSE in February 2017. We use observations from the remaining 11 lit exchanges in the regression analyses. Because Rule 605 reports include only dollar effective spreads, we calculate percentage effective spreads using monthly mean share prices.

Table 5 shows the regression results for each order size group. To save space, we report the results for the control variables in the Online Appendix (Table OA10). For the smallest order size group, we find that the coefficient on

$Q \times Event \times TBC$ is positive and significant and the coefficient on $Q \times Event$ is positive but insignificant, indicating that the pilot program results in a greater increase in effective spreads for the pilot stocks with larger *TBC* values (lower pre-pilot spreads). The coefficient on $TA \times Event$ is negative and significant, indicating that the *TA* rule results in a decrease in effective spreads for small orders. The coefficient on $Event \times TBC$ is positive and significant, indicating that the pilot program increases execution costs for the control stocks and this increase is larger for the control stocks with lower pre-pilot spreads. Overall these results are similar to the results for *EffectiveSpread* in Table 3.

For the 500–1999 order size group, the coefficients on $Q \times Event \times TBC$ and $Q \times Event$ are both insignificant, indicating that the pilot program has no effect on the Rule 605 effective spread for this group. The coefficient on $Event \times TBC$ is positive and significant, indicating that the pilot program increases execution costs for the control stocks with lower pre-pilot spreads.

For the second largest order size group (2000–4999), the coefficient on $Q \times Event \times TBC$ is negative and significant (−0.4390), and the coefficient on $Q \times Event$ is positive and significant (0.2394). For stocks with $TBC = 1$, the effect of the *Q* rule on Rule 605 effective spreads is −0.1996 (= 0.2394 − 0.4390). More generally, the *Q* rule results in a decrease in effective spreads for stocks with $[-0.4390 \times (Q \times Event \times TBC)] + [0.2394 \times (Q \times Event)] < 0$ or, alternatively, $TBC > 0.5453$, which corresponds to the 49th percentile value of *TBC* for the pilot stocks. These results indicate that, for order sizes between 2000 and 4999 shares, the *Q* rule results in a decrease in Rule 605 effective spreads for nearly 50% of the pilot stocks.

For the largest order size group (5000+), the coefficient on $Q \times Event \times TBC$ is negative and significant (−1.2298), and the coefficient on $Q \times Event$ is positive and significant (0.8975). For stocks with $TBC = 1$, the effect of the *Q* rule on Rule 605 effective spreads is −0.3323 (= 0.8975 − 1.2298). The *Q* rule results in a decrease in Rule 605 effective spreads for stocks with $[-1.2298 \times (Q \times Event \times TBC)] + [0.8975 \times (Q \times Event)] < 0$ or, alternatively, $TBC > 0.7298$.

To summarize, the pilot program results in an increase in Rule 605 effective spreads for small orders, but a decrease in Rule 605 effective spreads for many large orders. These results are generally consistent with the results for small orders in Table 3 and the results for large orders in Table 4.

6. Additional results

This section explores the effect of the pilot program on the control stocks, whether the effect of the pilot program extends beyond the top of the book, and whether our results are driven by phantom liquidity.

6.1. Effects of the pilot program on the control stocks

In Section 4.2, we extend the spillover effect analysis of Rindi and Werner (2017) by looking at the spillover

effect of the pilot program for large orders. In this section, we interpret the results in Table 3 and Table 4 to shed further light on the spillover effect. We explore whether the spillover effect varies with the pre-pilot level of liquidity for the control stocks. Table 3 shows that the coefficients on $Event \times TBC$ in the quoted, effective, and realized spread and price impact regressions are all positive and significant. These results indicate that the pilot program results in a larger increase in spreads and price impact for the control stocks with lower spreads (i.e., larger TBC values) before the pilot program. The coefficients on $Event \times TBC$ are insignificant in the depth regression, indicating that the pilot program has no effect on the NBBO depth for the control stocks as a whole.

Table 4 shows that the coefficient on $Event \times TBC$ in the $CumuDepth$ regression is not significant, indicating that the pilot program has no spillover effect on $CumuDepth$ for the control stocks. The coefficient on $Event \times TBC$ is negative and significant in the $PAVAIL$ regression, indicating that the pilot program results in a greater decrease in the chance of the limit order book displaying at least five thousand shares on both the ask and bid side of the book for the control stocks with lower pre-pilot spreads. The coefficients on $Event \times TBC$ are positive and significant in the regression models of $PCRT500$ and $CumuPriceImpact$, suggesting that the pilot program results in a larger increase in $PCRT500$ and $CumuPriceImpact$ for the control stocks with lower pre-pilot spreads. The coefficient on $Event \times TBC$ is not significant in the regression model of $PCRT5000$, indicating that the effect of the pilot program on $PCRT5000$ (i.e., the pilot program increases the execution cost of large orders) for the control stocks shown in Table 2 does not vary with their pre-pilot spreads.

To summarize, the results in this and Sections 5.1 and 5.2 suggest that the pilot program (1) decreases the liquidity for small orders for the pilot stocks and this decrease is larger for the pilot stocks with lower pre-pilot spreads (i.e., larger TBC values), (2) increases the liquidity for large orders for the pilot stocks and this increase is also larger for the pilot stocks with lower pre-pilot spreads, (3) decreases the liquidity for small orders for the control stocks and this decrease is larger for the control stocks with lower pre-pilot spreads, and (4) decreases the liquidity for large orders for the control stocks.

Results (1) and (2) jointly imply a change in the relative execution cost of small and large orders for the pilot stocks, especially those with low pre-pilot spreads. Results (1) and (3) jointly imply an illiquidity spillover (e.g., an increase in spreads) from the pilot stocks to the control stocks for small orders, especially between those stocks with low pre-pilot spreads. This result suggests that liquidity providers increase the spread of the control stocks with low pre-pilot spreads when they are forced to increase the spread of the pilot stocks with low pre-pilot spreads because of the binding tick size. Alternatively, the decrease in the liquidity of the control stocks could result from the migration of liquidity providers from the control stocks to the pilot stocks as they find it more profitable (e.g., larger spreads) to provide liquidity for the pilot stocks. Results (2) and (4) jointly imply liquidity migration from the control stocks to the pilot stocks for large orders. Results (3) and

(4) imply no liquidity spillover between small and large orders for the control stocks.

6.2. Effects of the pilot program beyond the top of the book

This section examines the effect of the Q , T , and TA rules on liquidity beyond the top of the book for the treatment and control stocks using regression model Eq. (8). The first two columns in Table 6 show the effects of the rules on $CumuDepth$ and $PAVAIL$, excluding quotes at the NBBO. We report the results for the control variables in the Online Appendix (Table OA11). We find that the coefficients on $Q \times Event$ are negative and significant and that the coefficients on $Q \times Event \times TBC$ are positive and significant in both regressions. These results are identical to those in Table 4, suggesting that the effect of the Q rule on liquidity extends beyond the top of the book. As in Table 4, the coefficient on $Event \times TBC$ is not significant in the $CumuDepth_{w/oNBBO}$ regression, suggesting no spillover effect. As in Table 4, the coefficient on $Event \times TBC$ is negative and significant in the $PAVAIL_{w/oNBBO}$ regression, confirming the larger spillover effect of the pilot program on $PAVAIL_{w/oNBBO}$ for the control stocks with lower pre-pilot spreads.

The last two columns in Table 6 show the results for $PCRT500$ and $PCRT5000$ when we estimate these variables after omitting NBBO quotes. The coefficients on $Q \times Event$ are not significantly different from zero in both regressions, indicating that the Q rule does not have an effect on the round-trip trading cost of five hundred and five thousand shares for the pilot stocks with large pre-pilot spreads (i.e., stocks with zero TBC) once we ignore its effect on the top of the book. The coefficients on $Q \times Event \times TBC$ have different signs between the two models. For $PCRT500_{w/oNBBO}$, the coefficient is positive and significant, suggesting that the cost of trading five hundred shares increases with TBC once we ignore the effect of the Q rule on the top of the book. The coefficient on $Q \times Event \times TBC$ for $PCRT5000_{w/oNBBO}$ is negative and significant, suggesting that the roundtrip trading cost of larger orders decreases with TBC even when we ignore the top of the book liquidity. The signs and statistical significance of the coefficients on $Event \times TBC$ are generally identical to those in Table 4, indicating that our inferences on spillover effects are robust to whether or not we include the top of the book liquidity.

6.3. Testing whether our results are driven by phantom liquidity

Our finding that the pilot program decreases the execution cost of large orders is at odds with the view expressed by some buy-side traders that it has had a negative impact on execution costs. For example, Barron's (2018) concludes that congress' failed stock market experiment cost investors \$900 million. Despite some bad publicity, the SEC (2018, p. 41) concludes that "[t]he Pilot appears to have increased liquidity". The change in the tick size provides a wealth transfer. Hence, it is important to analyze the overall impact of the pilot. Furthermore, not all market observers associate the pilot program with larger execution

Table 6

Regression results using liquidity measures that are constructed after omitting national best bid and offer (NBBO) quotes.

This table examines the effect of the *Q*, *T*, and *TA* rules on liquidity beyond the top of the book for the treatment and control stocks. We reconstruct liquidity measures in Table 4 after omitting NBBO quotes and reestimate the regression models. We report the results for the control variables in the Online Appendix (Table OA.11). Figures in parentheses are *t*-statistics. Standard errors are clustered by both firm and day. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *N* denotes the number of observations. See Tables 1 and 4 for the definition of each variable.

Variable	CumuDepth _{w/oNBBO}	PAVAIL _{w/oNBBO}	PCRT500 _{w/oNBBO}	PCRT5000 _{w/oNBBO}
$Q \times \text{Event} \times \text{TBC}$	9.8151*** (3.72)	0.2697*** (9.99)	1.3392*** (3.68)	-2.2229*** (-3.03)
$Q \times \text{Event}$	-2.3494** (-2.02)	-0.0602*** (-4.65)	-0.1499 (-0.53)	-0.4773 (-0.98)
$\text{Event} \times \text{TBC}$	1.3596 (0.70)	-0.0268** (-2.18)	0.6796** (2.34)	0.1033 (0.22)
$T \times \text{Event}$	0.2451 (0.35)	-0.0062 (-0.36)	0.1190 (0.58)	-0.1675 (-0.35)
$TA \times \text{Event}$	1.6581 (1.22)	0.0133 (0.74)	-0.1192 (-0.60)	-0.1998 (-0.40)
Control variables	Yes	Yes	Yes	Yes
Stock and day fixed effects	Yes	Yes	Yes	Yes
<i>N</i>	777,589	777,589	777,588	777,588
Adjusted R^2	0.3854	0.6486	0.8540	0.8684
<i>F</i> -statistic	23.45	18.45	57.30	28.53

costs. For example, Larry Tabb, the research chairman of TABB Group, writes, “All investors are paying for the largest investors to trade in greater size with less market movement. This market is worse for smaller orders and better for larger ones; however, all are paying more to trade this sector, and while this should coax more investors into the small cap pool, we just aren’t seeing it yet” (Tabb, 2016, para. 16).

In this section, we explore a possible explanation for the inconsistency: the duplication of liquidity across different markets that arises from the activities of high-frequency traders. van Kervel (2015) finds significant cancellations of orders in competing markets after execution of an order. The author interprets this finding as evidence that high-frequency traders adjust order placement after a trade. We explore whether the higher available liquidity for large orders during the pilot period is due to a failure to consider canceled orders following an execution (thus, overestimating the available liquidity). Although directly addressing this issue is difficult due to data limitations, we conduct a test to address this possibility at least partially.

Lee and Watts (2018) show that the tick pilot program results in a decrease in order cancellation rates and interpret this result as evidence that larger tick sizes reduce algorithmic trading (AT). To the extent that phantom liquidity is linked to AT (e.g., AT can quickly and easily cancel orders that were not affected by a trade) and order cancellation rates increase with AT, we can shed some light on whether the increase in the liquidity for large orders is due to phantom liquidity by analyzing the effect of the *Q*, *T*, and *TA* rules on order cancellation rates. Following Hasbrouck and Saar (2013) and Lee and Watts (2018), we measure order cancellation rates by the cancel-to-trade ratios. We obtain the total number of messages that are canceled across all exchanges from the SEC. We obtain the total number of trades (*Trades*) during the regular trading

hours on a given day from the Daily TAQ database. The order cancellation rate, *Cancellations*, is the ratio of the number of cancellations to *Trades*.

Table 7 shows the regression results. The first column shows the results when we do not include *TBC* and its interactions with other variables in the regression model. The second column shows the results when we include these variables. To save space, we report the results for the control variables in the Online Appendix (Table OA12). The negative and significant coefficients on $Q \times \text{Event}$ in both regressions together with the insignificant coefficient on $Q \times \text{Event} \times \text{TBC}$ indicate that the *Q* rule decreases the order cancellation rate for the pilot stocks more than the control stocks and the effect of the *Q* rule on the order cancellation rate for the pilot stocks does not vary with their *TBC* values. The decrease in order cancellation rates is consistent with the SEC (2018) assessment of the pilot program. We interpret these results as partial evidence that the positive impact of the *Q* rule on the liquidity for large orders shown earlier are perhaps not attributed to phantom (ghost) liquidity. The second column shows that the coefficient on $\text{Event} \times \text{TBC}$ is positive and significant, suggesting higher order cancellation rates for the control stocks with lower pre-pilot spreads. The coefficients on $T \times \text{Event}$ and $TA \times \text{Event}$ are not significantly different from zero, indicating that both the trade and trade-at rules do not have an impact on order cancellation rates.

van Kervel (2015) reports that transactions on the three most active trading venues are followed by significant cancellations of limit orders (i.e., 38% to 84% of the transaction size) on the same side of competing venues’ order books within one second. To further address the phantom liquidity concern, we estimate regression model Eqs. (7) and (8) using the values of *CumuDepth*, *PCRT*, and *PAVAIL* measured one second after each trade. We find results that are qualitatively similar to those reported in Table 4. We provide these results in the Online Appendix (Table OA13).

Table 7

Effects of the quote (Q), trade (T), and trade-at (TA) rules on order cancellation rate.

This table shows the results of the following regression model: $Cancellations_{i,d} = \beta_1 Q_i \times Event_d \times TBC_{i,d} + \beta_2 Q_i \times Event_d + \beta_3 Event_d \times TBC_{i,d} + \beta_4 T_i \times Event_d + \beta_5 TA_i \times Event_d + Control\ Variables_{i,d} + \lambda_i + \theta_d + \varepsilon_{i,d}$, where $Cancellations_{i,d}$ is the number of cancellations scaled by the number of trades, Q_i is the quote rule dummy variable (which is equal to one for stocks that are quoted in \$0.05 increments and zero otherwise), T_i is the trade rule dummy variable (which is equal to one for stocks that are traded in \$0.05 increments and zero otherwise), TA_i is the trade-at rule dummy variable (which is equal to one for stocks that are subject to the trade-at requirement and zero otherwise), $Event_d$ is the tick pilot dummy variable (which is equal to one for the tick pilot period and zero otherwise), $TBC_{i,d}$ is the extent to which the new quote tick size (\$0.05) is a binding constraint on quoted spreads in the pilot period (measured by the percentage of time during which quoted spreads are equal to or smaller than 5 cents in the pre-pilot period), λ_i denotes stock fixed effects, and θ_d denotes day fixed effects. We include share price, market capitalization (in log), and return volatility as control variables. We report the results for the control variables in the Online Appendix (Table OA.12). Figures in parentheses are *t*-statistics. Standard errors are clustered by both firm and day. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *N* denotes the number of observations.

Variable	Cancellations (1)	Cancellations (2)
$Q \times Event \times TBC$		-2.6389 (-1.47)
$Q \times Event$	-4.7913*** (-7.46)	-3.2006** (-2.34)
$Event \times TBC$		11.139*** (9.03)
$T \times Event$	-0.3673 (-0.48)	-0.3889 (-0.54)
$TA \times Event$	2.0993 (1.54)	2.0898 (1.56)
Control variables	Yes	Yes
Stock and day fixed effects	Yes	Yes
<i>N</i>	805,347	805,347
Adjusted R^2	0.0318	0.0325
F-statistic	19.98	40.75

7. Effects of the pilot program on information flow, price informativeness, and trading activity

Previous sections show that the pilot program results in a decrease in the execution cost of large orders and an increase in the execution cost of small orders. These changes could lead to changes in trading behavior and price discovery. For instance, traders can use more large orders and fewer small orders than before. To the extent that the information content of an order varies with its size, the pilot program could have caused certain changes in information flow and pricing efficiency.

7.1. Return volatility and information flow

We use *FundamentalVolatility* as a proxy for how much information regarding fundamentals is impounded into prices, which is the difference between total volatility (*TotalVolatility*) and the volatility of pricing error. We measure the volatility of pricing error by the standard deviation of intraday pricing errors (Hasbrouck, 1993). In addition, we examine whether the pilot program increases

or decreases the amount of firm-specific information incorporated into prices. Following Chen et al. (2007) and Morck et al. (2013), we measure firm-specific information by $1 - R^2$, where R^2 is obtained by regressing daily stock excess return (from the CRSP database) on Fama and French (1992) three factors (equity premium, return difference between small and large stocks, and return difference between high and low book-to-market stocks) for each quarter. We obtain daily factor data from Kenneth French's website: <https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>. A higher value of $1 - R^2$ indicates that more firm-specific information is incorporated into prices. Table 1 provides descriptive statistics of these variables.

The first two columns of Panel A of Table 8 show the aggregate effect of the pilot program on these variables. Panel A reports the regression results without the fixed effects to show the effect of the pilot program on the control stocks (i.e., the coefficients on *Event*). The coefficients on *Pilot* $\times$ *Event* are positive and significant, indicating that the pilot stocks have higher *FundamentalVolatility* and $1 - R^2$ compared with the control stocks in the pilot period. Panel B shows the effects of the Q and T rules on information flow from regressions with the fixed effects. We provide the results for the control variables with and without the fixed effects in the Online Appendix (Table OA14). The coefficient on $Q \times Event$ is negative and significant in the regression model of *FundamentalVolatility*, indicating that stocks with zero *TBC* (i.e., stocks with large equilibrium spreads) become less volatile after the implementation of the Q rule. In contrast, the coefficient on $Q \times Event \times TBC$ is positive and significant. The size of the coefficient in the first column suggests that the Q rule increases *FundamentalVolatility* for stocks with *TBC* values larger than 0.4444 ($= 0.0028 / 0.0063$), which corresponds to the 40th percentile value of *TBC* for the pilot stocks. Hence, the Q rule increases *FundamentalVolatility* for more than 60% of the pilot stocks. The Q rule improves information flow for the majority of pilot stocks, which is consistent with one of the main objectives of the Tick Size Pilot. In contrast, the coefficient on $T \times Event$ is not significantly different from zero, indicating that the trade rule has no effect on information flow. The coefficients on both $Q \times Event$ and $Q \times Event \times TBC$ are insignificant in the regression model of $1 - R^2$, suggesting that the effect of the Q rule on $1 - R^2$ does not vary with *TBC*.

The coefficient on *Event* is positive and significant in the *FundamentalVolatility* regression in Panel A, and the corresponding coefficient on $Event \times TBC$ is negative and significant in Panel B. These results suggest that the tick pilot program generally increases information flow for the non pilot stocks and this increase is smaller for stocks with higher *TBC* values (i.e., lower pre-pilot spreads).

7.2. Pricing efficiency

This section analyzes the effect of the pilot program on pricing efficiency using return autocorrelation and return predictability of order imbalance. We measure return autocorrelation (*Autocorrelation*) by the R^2 from the regression of $Return_t$ on $Return_{t-1}$, where $Return_t$ is the quote

Table 8

Effects of the pilot program on return volatility, pricing efficiency, and trading activity.

Panel A shows the results of the regression model $VAR_{i,d} = \beta_1 Pilot_i \times Event_d + \beta_2 Event_d + \beta_3 Pilot_i + Control\ Variables_{i,d} + \varepsilon_{i,d}$, and Panel B shows the results of the regression model $VAR_{i,d} = \beta_1 Q_i \times Event_d \times TBC_{i,d} + \beta_2 Q_i \times Event_d + \beta_3 Event_d \times TBC_{i,d} + \beta_4 T_i \times Event_d + \beta_5 TA_i \times Event_d + Control\ Variables_{i,d} + \lambda_i + \theta_d + \varepsilon_{i,d}$, where $VAR_{i,d}$ represents *FundamentalVolatility*, $1 - R^2$, *Autocorrelation*, *OIB_Predictability*, *DollarVolume*, *TradeSize*, or *Trades*. $Pilot_i$ is equal to one for the pilot stocks and zero otherwise and $Event_d$ is the tick pilot dummy variable (which is equal to one for the tick pilot period and zero otherwise). Q_i is the quote rule dummy variable (which is equal to one for stocks that are quoted in \$0.05 increments and zero otherwise), T_i is the trade rule dummy variable (which is equal to one for stocks that are subject to the trade-at requirement and zero otherwise), $TBC_{i,d}$ is the extent to which the new quote tick size (\$0.05) is a binding constraint on quoted spreads in the pilot period (measured by the percentage of time during which quoted spreads are equal to or smaller than 5 cents in the pre-pilot period), λ_i denotes stock fixed effects, and θ_d denotes day fixed effects. We include Chicago Board Options Exchange Volatility Index (*VIX*), share price, market capitalization (in log), dollar trading volume (in log), and return volatility as control variables. We use the logistically transformed value of $1 - R^2$, $\text{Log}[(1 - R^2) / (1 - (1 - R^2))]$, in the regression. Similarly, we use the logistically transformed values of *Autocorrelation* and *OIB_Predictability*. We report the results for the control variables in the Online Appendix (Table OA.14, Panel A and Panel B). Panel C of Table OA.14 shows the aggregate effects of the pilot program from regressions with the fixed effects. Panel D of Table OA.14 shows the rule effects from regressions without the fixed effects. Figures in parentheses are *t*-statistics. Standard errors are clustered by both firm and day. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively. *N* denotes the number of observations. See Table 1 for the definition of each dependent variable.

	Fundamental-Volatility (1)	$1 - R^2$ (2)	Autocorrelation (3)	OIB_Predictability (4)	DollarVolume (5)	TradeSize (6)	Trades (7)
Panel A: Aggregate effects of the pilot program (regression results without the fixed effects)							
<i>Pilot</i> × <i>Event</i>	0.0009*** (3.08)	0.0842*** (4.71)	-0.1977*** (-7.52)	-0.2401*** (-13.70)	-0.0822*** (-2.89)	160.15*** (5.84)	-344.36*** (-6.77)
<i>Pilot</i>	-0.0008** (-1.97)	0.0086 (0.33)	0.0225 (1.18)	0.0092 (0.42)	0.0187 (0.52)	52.9673 (1.42)	87.528 (1.44)
<i>Event</i>	0.0008** (2.36)	-0.1058 (-0.63)	-0.0628*** (-3.69)	-0.0074 (-0.44)	0.1232*** (5.20)	-86.153*** (-3.31)	310.34*** (7.67)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	634,032	13,440	787,617	714,415	805,766	805,766	805,766
Adjusted R^2	0.2426	0.3286	0.0429	0.1181	0.6845	0.5641	0.2731
<i>F</i> -statistic	191.72	264.75	254.29	866.38	1197.04	921.43	201.91
Panel B: Rule effects (regression results with the fixed effects)							
<i>Q</i> × <i>Event</i> × <i>TBC</i>	0.0063*** (7.09)	0.1348 (1.31)	-0.0742 (-0.82)	-0.2392*** (-4.53)	-0.0395 (-0.47)	126.29 (1.12)	-870.94*** (-5.97)
<i>Q</i> × <i>Event</i>	-0.0028*** (-4.26)	0.0418 (0.62)	-0.1370** (-2.50)	-0.0996** (-2.56)	-0.0656 (-1.10)	119.47 (1.50)	160.42** (2.08)
<i>Event</i> × <i>TBC</i>	-0.0024*** (-3.62)	0.346*** (5.07)	0.1250*** (2.88)	0.3525*** (9.34)	0.0806 (1.28)	543.17*** (7.45)	446.00*** (3.64)
<i>T</i> × <i>Event</i>	-0.0001 (-0.24)	-0.0734 (-1.31)	-0.0391 (-0.68)	0.0057 (0.19)	0.0603 (1.31)	52.062 (1.50)	-11.160 (-0.15)
<i>TA</i> × <i>Event</i>	0.0007 (1.55)	0.0536 (0.93)	0.0071 (0.13)	-0.0389 (-1.29)	-0.0427 (-0.85)	-167.83*** (-3.33)	35.541 (0.43)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock and day fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	634,032	13,440	787,617	714,415	805,766	805,766	805,766
Adjusted R^2	0.5089	0.5625	0.0731	0.1480	0.8556	0.6812	0.4991
<i>F</i> -statistic	288.71	31.44	61.38	528.24	220.15	96.67	65.29

midpoint return from minute $t - 1$ to minute t . We measure return predictability of order imbalance (*OIB_Predictability*) by the R^2 from regressing one-minute stock return on the order imbalance in the previous one-minute interval, with order imbalance being the difference in dollar trading volume between buyer- and seller-initiated trades. Chordia et al. (2005) show that order imbalances can predict future returns over short time horizons. We interpret high *OIB_Predictability* as evidence of lower efficiency. Table 1 provides descriptive statistics of *Autocorrelation* and *OIB_Predictability*.

The third and fourth columns of Panel A of Table 8 show the aggregate effect of the pilot program on these variables. The coefficients on *Pilot* × *Event* are negative and significant, indicating that the pilot stocks have lower *Autocorrelation* and *OIB_Predictability* compared with the control stocks. Panel B shows that the coefficients

on *Q* × *Event* are negative and significant regardless of whether the dependent variable is *Autocorrelation* or *OIB_Predictability*. This result is consistent with the finding of Comerton-Forde et al. (2019) that the *Q* rule decreases autocorrelations in 15-second quote midpoint returns. While *TBC* does not have any significant incremental effect on *Autocorrelation*, it does have a negative and significant effect on *OIB_Predictability*. In other words, stocks with higher *TBC* values have experienced even further increases in pricing efficiency (i.e., returns are less predictable) due to the larger quote tick size.

Both the trade and trade-at rules do not have any significant effect on either dependent variable. Comerton-Forde et al. (2019) suggest two channels through which the *TA* rule can affect pricing efficiency. The *TA* rule increases pricing efficiency by increasing trading on inverted fee venues and decreases pricing efficiency by decreasing

trading on dark pools. Hence, the net effect of the *TA* rule on pricing efficiency depends on relative strength of these channels. We interpret the insignificant coefficients on $TA \times Event$ as evidence that these channels are equally strong.

Panel A shows that the coefficient on *Event* is negative and significant, and Panel B shows that the coefficients on $Event \times TBC$ are positive and significant in the *Autocorrelation* regressions. These results suggest that although the pilot program results in a greater increase in *Autocorrelation* for the control stocks with larger *TBC* values, the negative effect of the pilot program on the autocorrelation of some control stocks dominates the positive effect of the pilot program on the autocorrelation of other control stocks. Panel A shows that the coefficient on *Event* is insignificant, and Panel B shows that the coefficients on $Event \times TBC$ are positive and significant in the *OIB_Predictability* regressions. These results suggest that although the pilot program results in a greater increase in *OIB_Predictability* for the control stocks with larger *TBC* values, the negative effect of the pilot program on the *OIB_Predictability* of some control stocks is offset by the positive effect of the pilot program on the *OIB_Predictability* of other control stocks, resulting in a zero net effect on *OIB_Predictability* for the control stocks as a whole.

7.3. Effects of the pilot program on trading volume, number of trades, and trade size

Although prior research (see O'Hara, 2015) shows that high-frequency trading has improved market quality (e.g., lower spreads and faster execution speed), some market observers believe that low trading volume in stocks of small companies in recent years could be due, at least in part, to the proliferation of high-frequency trading in stocks of large companies. The IPO Task Force (2011) concluded that the US listing gap could be attributed at least in part to a shift in trading interest from long-term investing in emerging growth companies to high-frequency trading in large-cap stocks, which led to a decrease in initial public offerings. Another possible reason for low trading volume in stocks of small companies is that investors are increasingly more concerned about adverse selection risks associated with trading in these stocks as stock markets are more dominated by opportunistic traders who are equipped with advanced technologies. Because traders who do not have similar technologies cannot compete with these traders, they could find the market unfair and inequitable and shy away from it altogether, reducing trading volume. To the extent that a larger trade tick size increases the cost of front running, one can conjecture that the tick pilot program can increase trading activity by alleviating traders' concerns over opportunistic trading.

We measure trading activity by the average daily dollar trade volume (*DollarVolume*), total number of trades (*Trades*), and average dollar trade size (*TradeSize*). Panel A of Table 8 shows that the coefficients on $Pilot \times Event$ are negative and significant in the *DollarVolume* and *Trades* regressions and positive and significant in the *TradeSize* regression, indicating that the pilot program results in a decrease in the dollar trading volume and the number of trades (*Trades*), but an increase in the average trade size.

We show in Panel B of Table 2 that the coefficients on $Pilot \times Event$ in the *PCRT* regressions are all significant and negative, indicating a reduction in the execution cost of large orders for the pilot stocks. We also show that the coefficients on $Pilot \times Event$ in the *CumuDepth* and *PAVAIL* regressions are all significant and positive, indicating an increase in displayed liquidity for the pilot stocks. Overall, these results are consistent with our expectation that traders use larger orders as visible liquidity increases and the execution cost of large orders decreases in the pilot period.

Panel B of Table 8 shows that the effect of the larger quote and trade tick sizes on trading volume and trade size does not vary with *TBC*. However, we find that the *Q* rule decreases the number of trades for the pilot stocks with *TBC* values greater than 0.1842 ($= 160.42 / 870.94$), which corresponds to the 22nd percentile value of *TBC*. If the execution cost of large orders decreases due to larger depths prompted by the *TA* rule, traders can reduce order splitting, increasing the average trade size (Seppi, 1997). Contrary to this expectation, the coefficient on $TA \times Event$ in the regression model of *TradeSize* is negative and significant, indicating that the *TA* rule decreases the average trade size. Because the coefficient on $Pilot \times Event$ measures the total effect of the *Q*, *T*, and *TA* rules, the positive and significant coefficient on $Pilot \times Event$ in the *TradeSize* regression in Panel A indicates that the positive effect of the *Q* and *T* rules (i.e., $Q \times Event \times TBC$, $Q \times Event$, and $T \times Event$) on *TradeSize* is greater than the negative effect of the *TA* rule. Finally, the significant coefficients on *Event* in the *DollarVolume*, *TradeSize*, and *Trades* regression models in Panel A suggest that the pilot program has a significant spillover effect on these variables for the control stocks.

8. Summary and concluding remarks

The Tick Size Pilot Program entails three distinct rules, namely, the quote, trade, and trade-at rules. Presumably, these rules are likely to have different impacts on trading and order placement strategies and, as a result, on various liquidity metrics and order execution quality. For instance, both the quote and trade-at rules could be more relevant for quote-based liquidity metrics, while the trade rule could be more relevant for trade-based liquidity metrics and order execution quality. In this study, we develop and test a few hypotheses by exploring various implications of these rules for different measures of liquidity and execution quality. Our empirical results are largely consistent with these hypotheses with a few exceptions (e.g., the effect of the *T* rule on liquidity is weaker than our prediction, suggesting either that the trade tick size does not affect front running or that front running has no impact on liquidity provision).

More important, our study provides evidence that the pilot program reduces the execution cost of large orders. This result is at odds with the view of some buy-side traders that the pilot program has a negative impact on market liquidity. One possible explanation for this inconsistency could be that their view is largely formed based on both the execution of small orders, which account for a large fraction of orders in the high-frequency trading

environment, and the difficulty associated with accurately measuring the execution cost of large orders.

Motivated by initial concerns of the IPO Task Force that recent regulations constrained information in stocks of small companies, we also analyze the effect of the pilot program on trading activities and informational environment. We find evidence that trading on private information becomes cheaper, possibly because liquidity for large trades improves or because large orders become less predictable, incentives to acquire private information increase, and prices become more efficient, better reflecting firm-specific fundamentals. The Tick Size Pilot Program, therefore, seems to address initial concerns of the IPO Task Force at least for some pilot stocks. Whether more efficient prices will have an impact on the propensity to be listed, as proposed by the IPO Task Force, is still an open question.

Finally, our analysis of the Tick Size Pilot Program is likely to have some ramifications for the upcoming Transaction Fee Pilot Program because both programs involve transfers of profits, with potential consequences on who trades and, therefore, on pricing efficiency. In this sense, our results have implications beyond the tick size pilot.

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